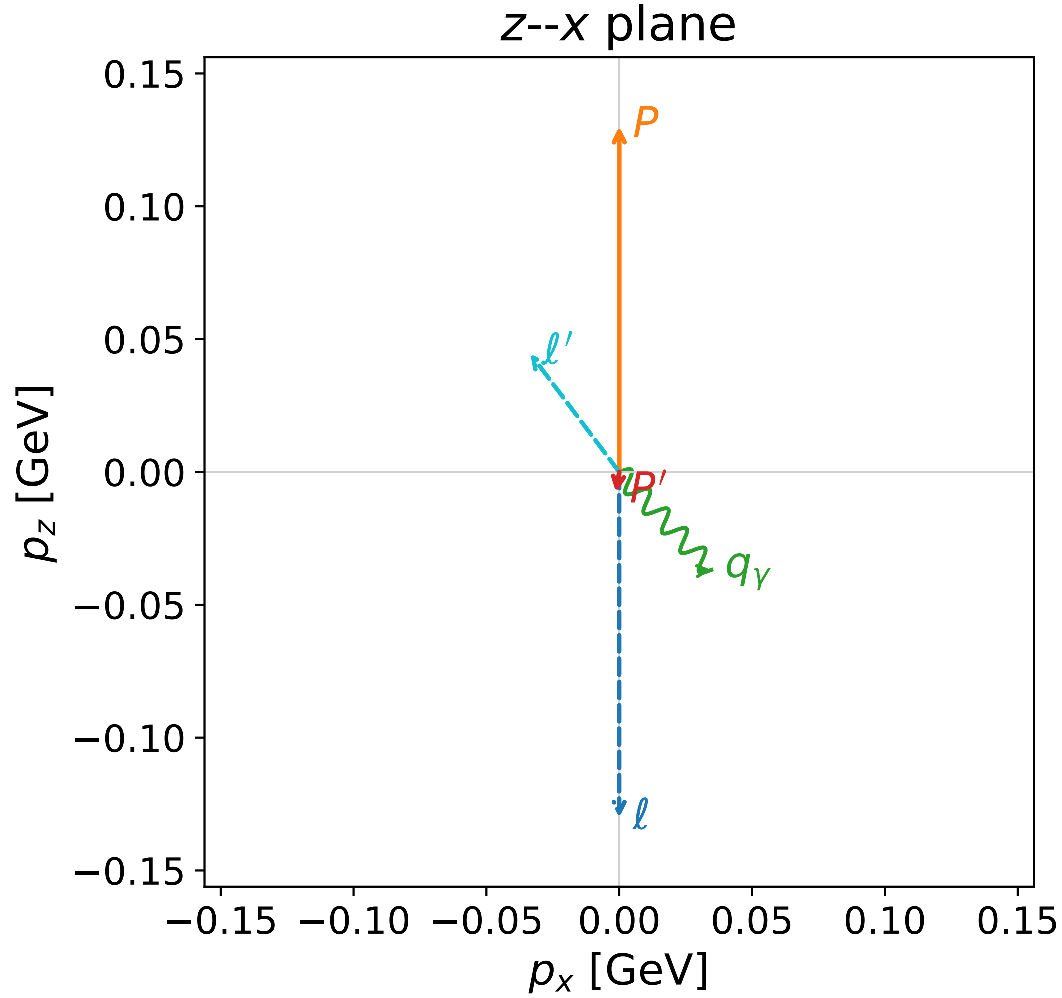
Canonical W configurations - cluster P6

```
species: electron
polarization cluster: P6
source cluster configurations: 82
canonical configurations collected: 19

canonical display criterion: exactly 3 retained final-state components
retained-component cutoff: fraction >= 2.0%
normalized amplitude display: bar length = |A_tilde|, bar color/text = phase

cluster retained-component count distribution:
  3 components: 19 configurations
  4 components: 23 configurations
  5 components: 9 configurations
  6 components: 28 configurations
  7 components: 3 configurations

CSV index:
  Output\GradientPhaseSpaceScan\electron\Data\W\dw\polarization_cluster_06\combined\cano
  nical_dw_polarization_cluster_06_configurations_electron.csv

Each following page is one complete momentum, kinematic-summary, and normalized-amplitude configuration.
```

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)




```
dw_mixing_angles_polarization_06_local_minimum_509
D_W=0.00259542
region: polarization_cluster_06_local_minimum_509
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0509
lepton mixing angle: alpha_e=2.3474721 rad
incoming lepton: -0.700912|+> +0.713247|->
proton mixing angle: alpha_p=1.3918247 rad
incoming proton: 0.178018|+> +0.984027|->
```

s=1.16007, $\sqrt{s}$=1.07706, $|\vec{P}|$=0.130085, $|\vec{P}'|$=0.00864853

$\theta_{p'}$=2.66104, θ_γ=2.13648, $\phi_{p'}$=1.87001, ϕ_γ=5.35198

E_γ=0.0687822, $\theta_{\ell'}$=0.884012, $\phi_{\ell'}$=2.23496

Q^2=0.0298612, x_B=0.188073, t=-0.0189123, W^2=1.00876, y=0.566603

four-momenta [E, p_x, p_y, p_z] GeV:

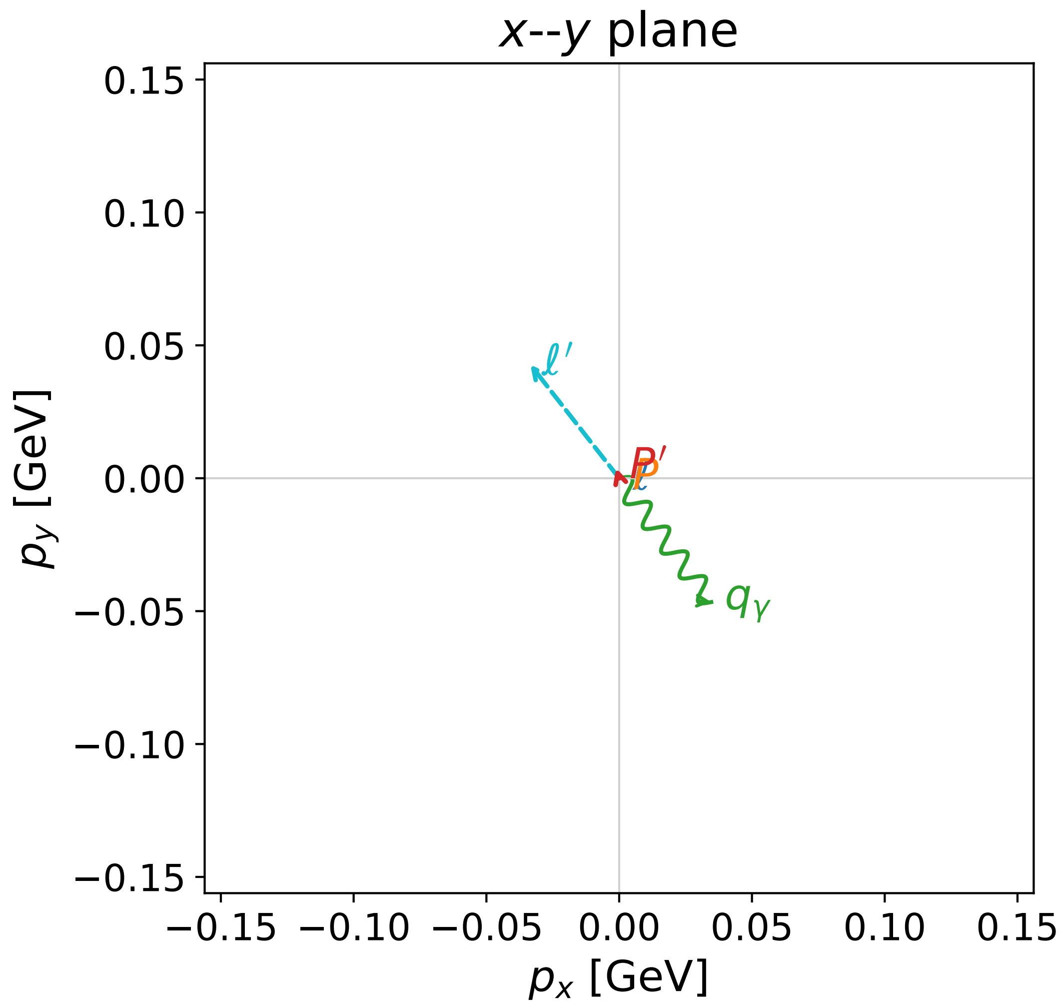
ℓ E= 0.1301 p_x= 0.0000 p_y= 0.0000 p_z= -0.1301

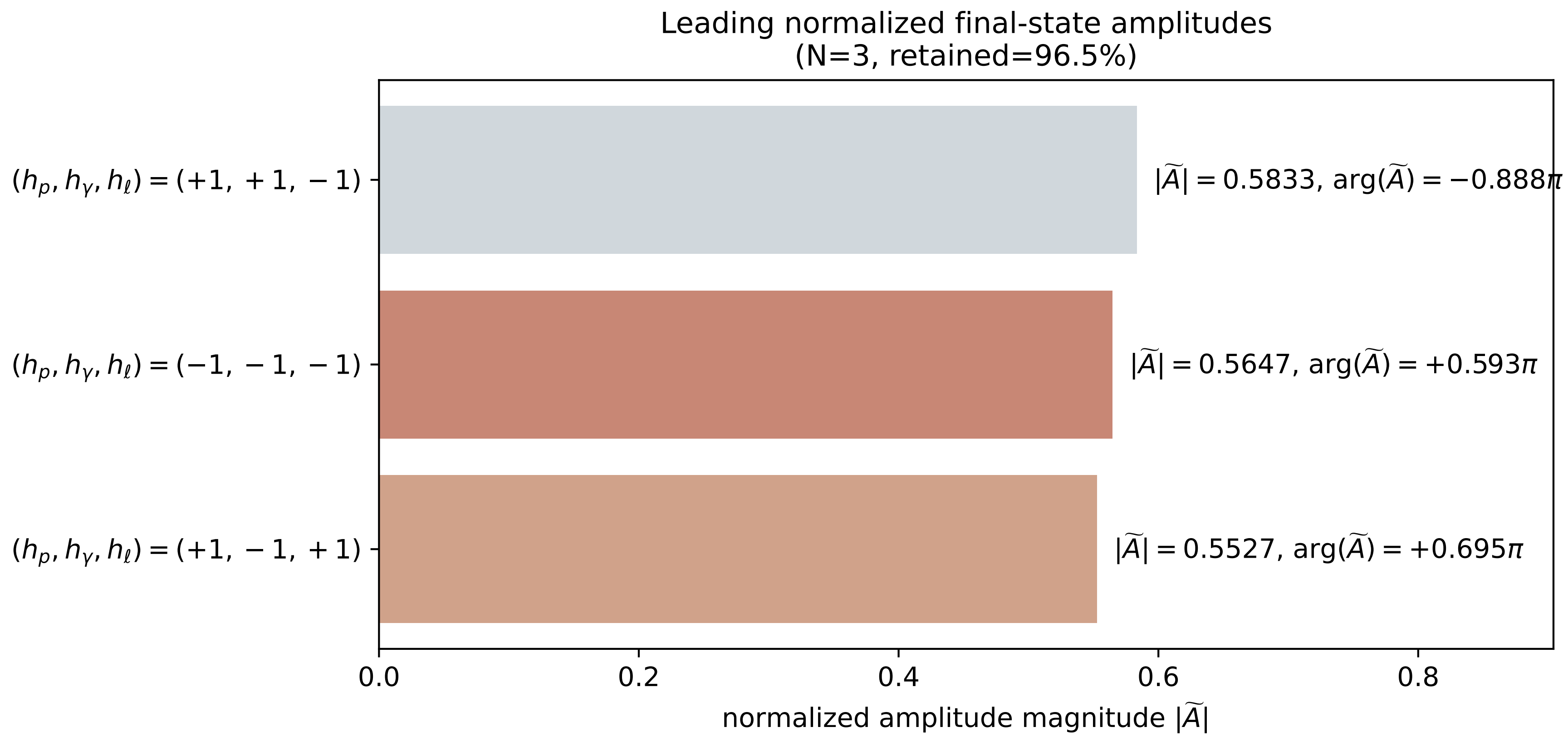
P E= 0.9470 p_x= 0.0000 p_y= 0.0000 p_z= 0.1301

ℓ' E= 0.0702 p_x= -0.0335 p_y= 0.0428 p_z= 0.0445

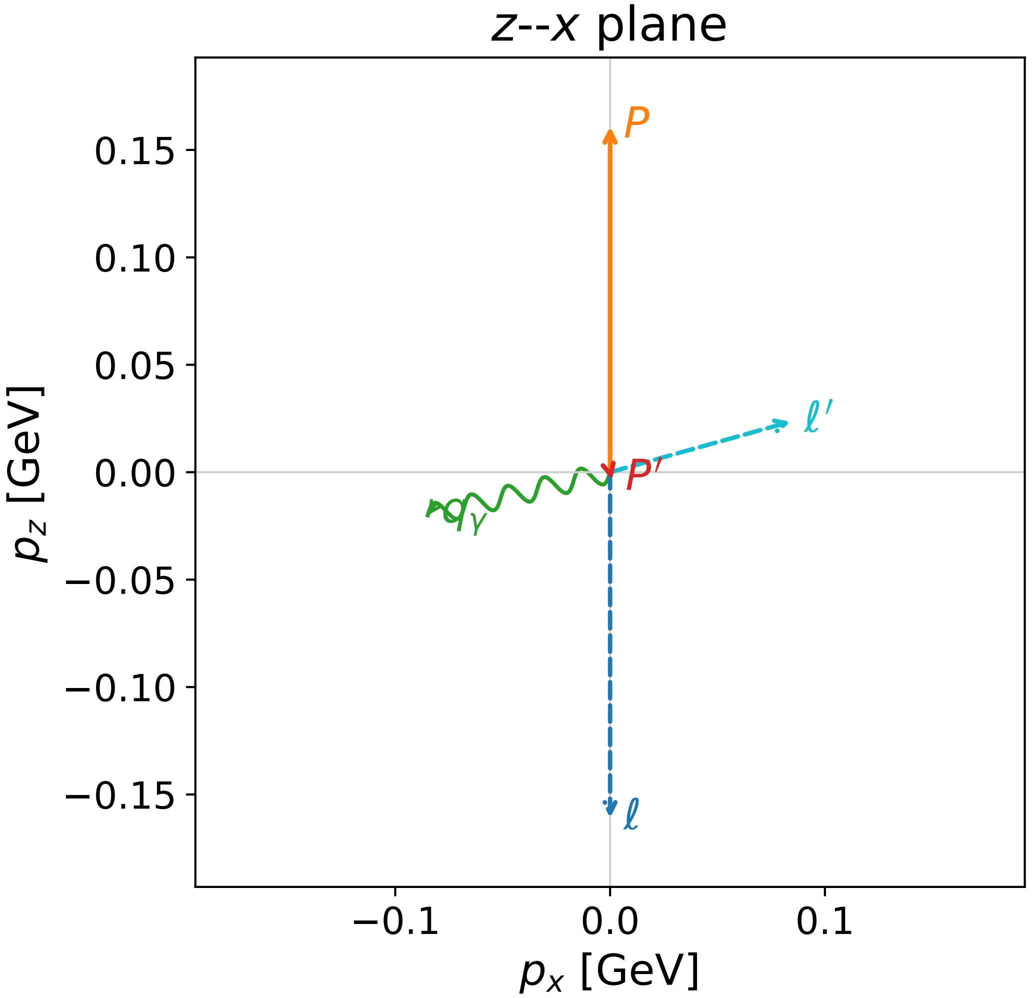
P' E= 0.9380 p_x= -0.0012 p_y= 0.0038 p_z= -0.0077

q_γ E= 0.0688 p_x= 0.0347 p_y= -0.0466 p_z= -0.0369





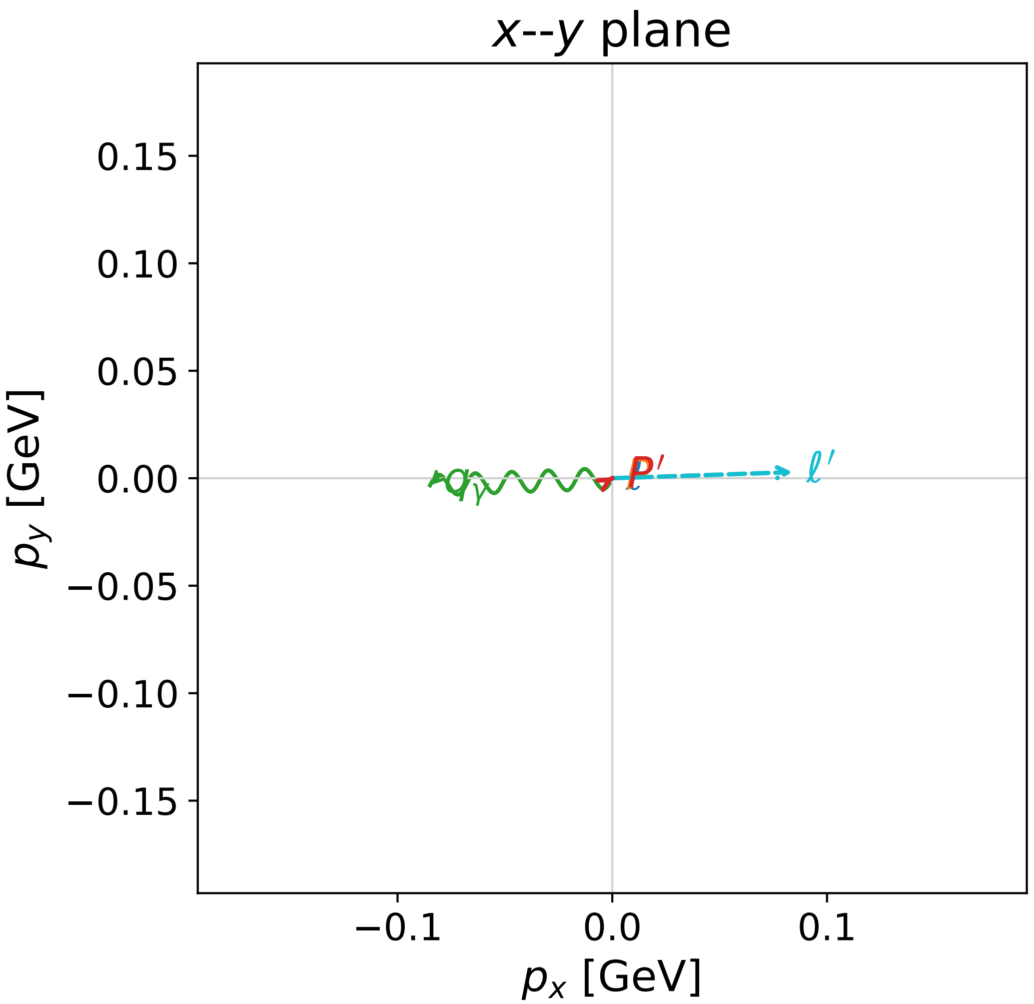
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

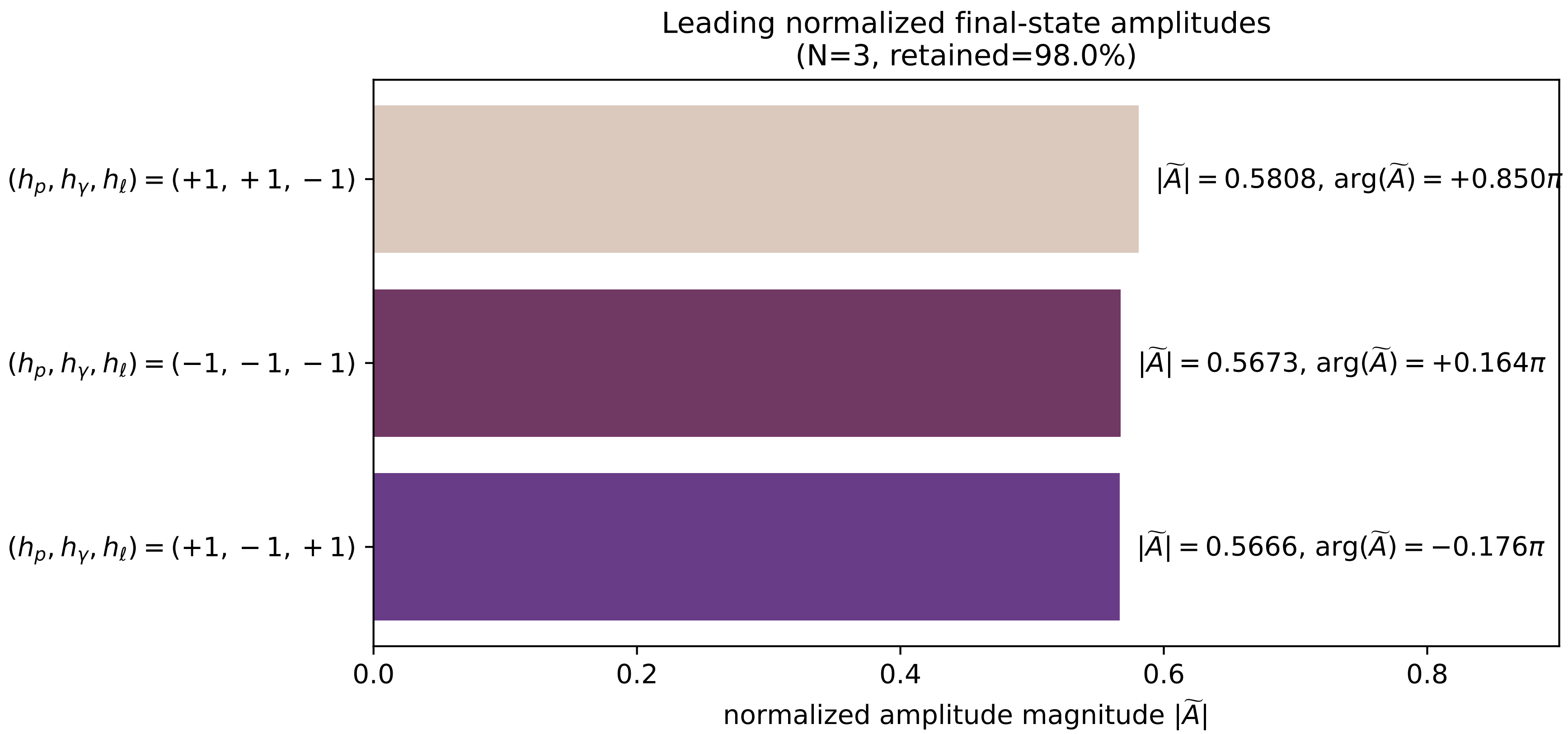


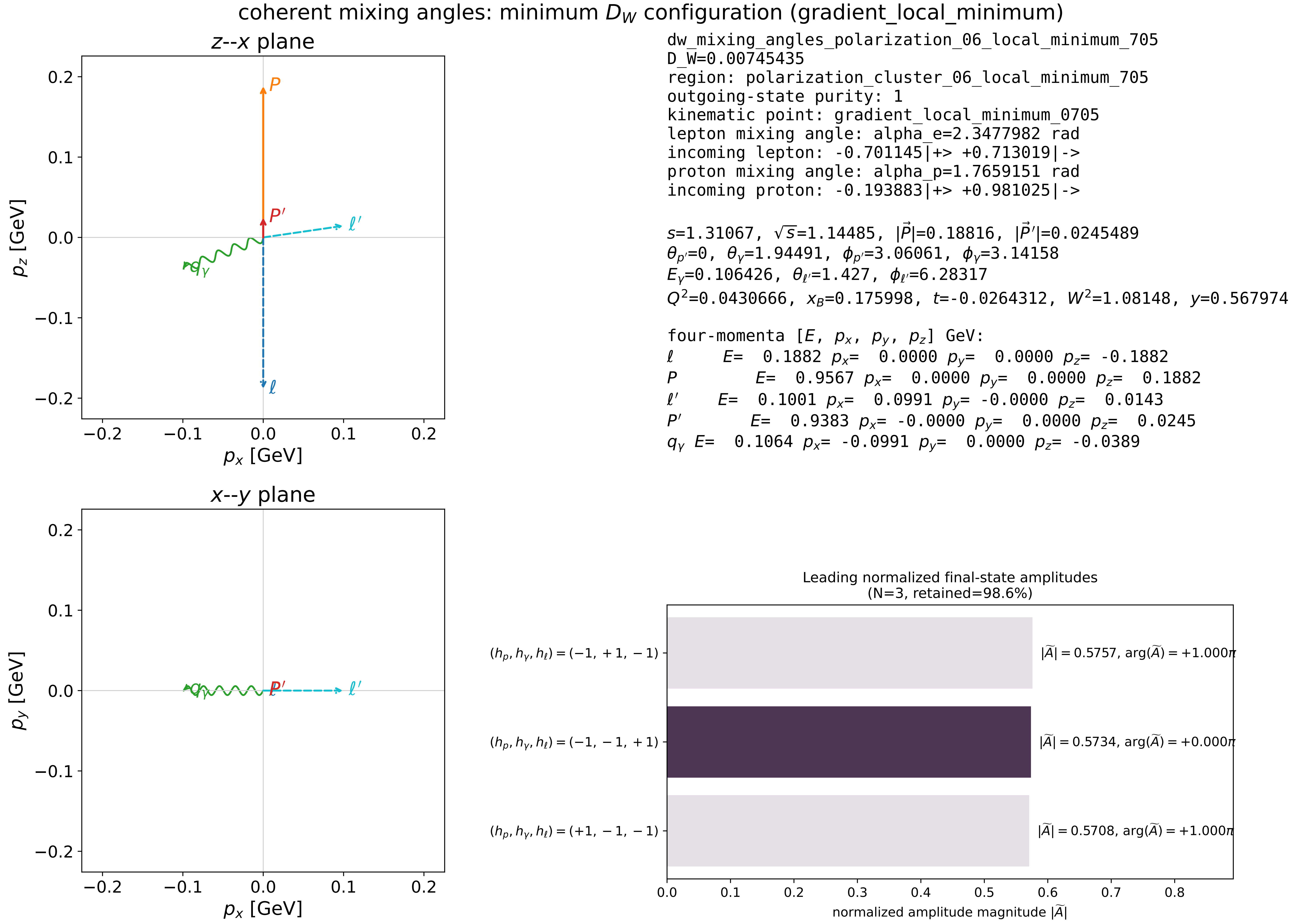
dw_mixing_angles_polarization_06_local_minimum_687
D_W=0.00662348
region: polarization_cluster_06_local_minimum_687
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0687
lepton mixing angle: alpha_e=2.3355152 rad
incoming lepton: -0.692334|+> +0.721577|->
proton mixing angle: alpha_p=1.7539808 rad
incoming proton: -0.182162|+> +0.983269|->

s=1.23777, $\sqrt{s}$=1.11255, $|\vec{P}|$=0.160858, $|\vec{P}'|$=0.003618
$\theta_{p'}$=2.87163, θ_γ=1.80235, $\phi_{p'}$=0.51582, ϕ_γ=3.18054
E_γ=0.0872511, $\theta_{\ell'}$=1.2981, $\phi_{\ell'}$=0.0336836
Q^2=0.035647, x_B=0.178828, t=-0.0268229, W^2=1.04353, y=0.556922

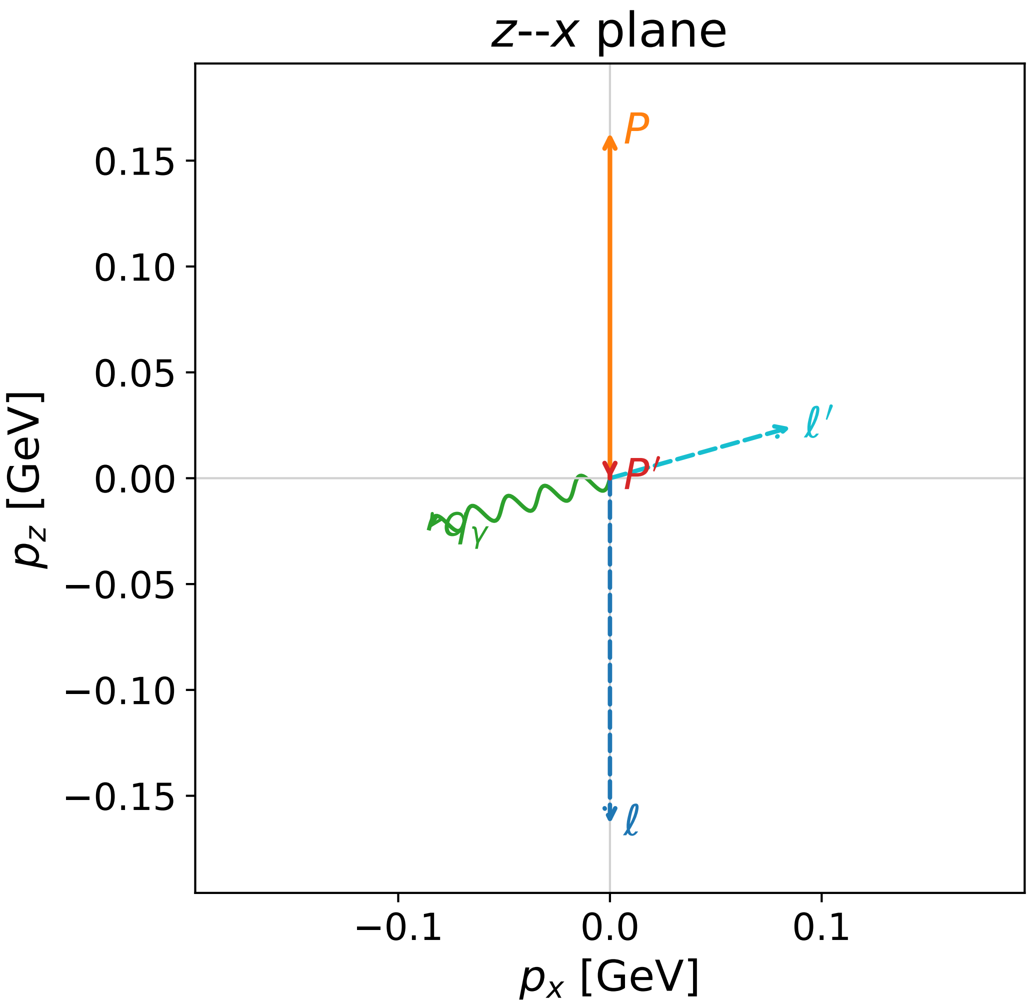
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1609 p_x= 0.0000 p_y= 0.0000 p_z= -0.1609
P E= 0.9517 p_x= 0.0000 p_y= 0.0000 p_z= 0.1609
ℓ' E= 0.0873 p_x= 0.0840 p_y= 0.0028 p_z= 0.0235
P' E= 0.9380 p_x= 0.0008 p_y= 0.0005 p_z= -0.0035
q_γ E= 0.0873 p_x= -0.0849 p_y= -0.0033 p_z= -0.0200






coherent mixing angles: minimum D_W configuration (gradient_local_minimum)
z--x plane
P
P'
ℓ'
q_γ
ℓ
p_z [GeV]
p_x [GeV]
x--y plane
p_y [GeV]
p_x [GeV]
dw_mixing_angles_polarization_06_local_minimum_705
D_W=0.00745435
region: polarization_cluster_06_local_minimum_705
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0705
lepton mixing angle: alpha_e=2.3477982 rad
incoming lepton: -0.701145|+> +0.713019|->
proton mixing angle: alpha_p=1.7659151 rad
incoming proton: -0.193883|+> +0.981025|->
s=1.31067, $\sqrt{s}$=1.14485, $|\vec{P}|$=0.18816, $|\vec{P}'|$=0.0245489
$\theta_{p'}$=0, θ_γ=1.94491, $\phi_{p'}$=3.06061, ϕ_γ=3.14158
E_γ=0.106426, $\theta_{\ell'}$=1.427, $\phi_{\ell'}$=6.28317
Q^2=0.0430666, x_B=0.175998, t=-0.0264312, W^2=1.08148, y=0.567974
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1882 p_x= 0.0000 p_y= 0.0000 p_z= -0.1882
P E= 0.9567 p_x= 0.0000 p_y= 0.0000 p_z= 0.1882
ℓ' E= 0.1001 p_x= 0.0991 p_y= -0.0000 p_z= 0.0143
P' E= 0.9383 p_x= -0.0000 p_y= 0.0000 p_z= 0.0245
q_γ E= 0.1064 p_x= -0.0991 p_y= 0.0000 p_z= -0.0389
Leading normalized final-state amplitudes
(N=3, retained=98.6%)
$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$
$|\widetilde{A}| = 0.5757$, $\arg(\widetilde{A}) = +1.000\pi$
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$
$|\widetilde{A}| = 0.5734$, $\arg(\widetilde{A}) = +0.000\pi$
$(h_p, h_\gamma, h_\ell) = (+1, -1, -1)$
$|\widetilde{A}| = 0.5708$, $\arg(\widetilde{A}) = +1.000\pi$
normalized amplitude magnitude $|\widetilde{A}|$

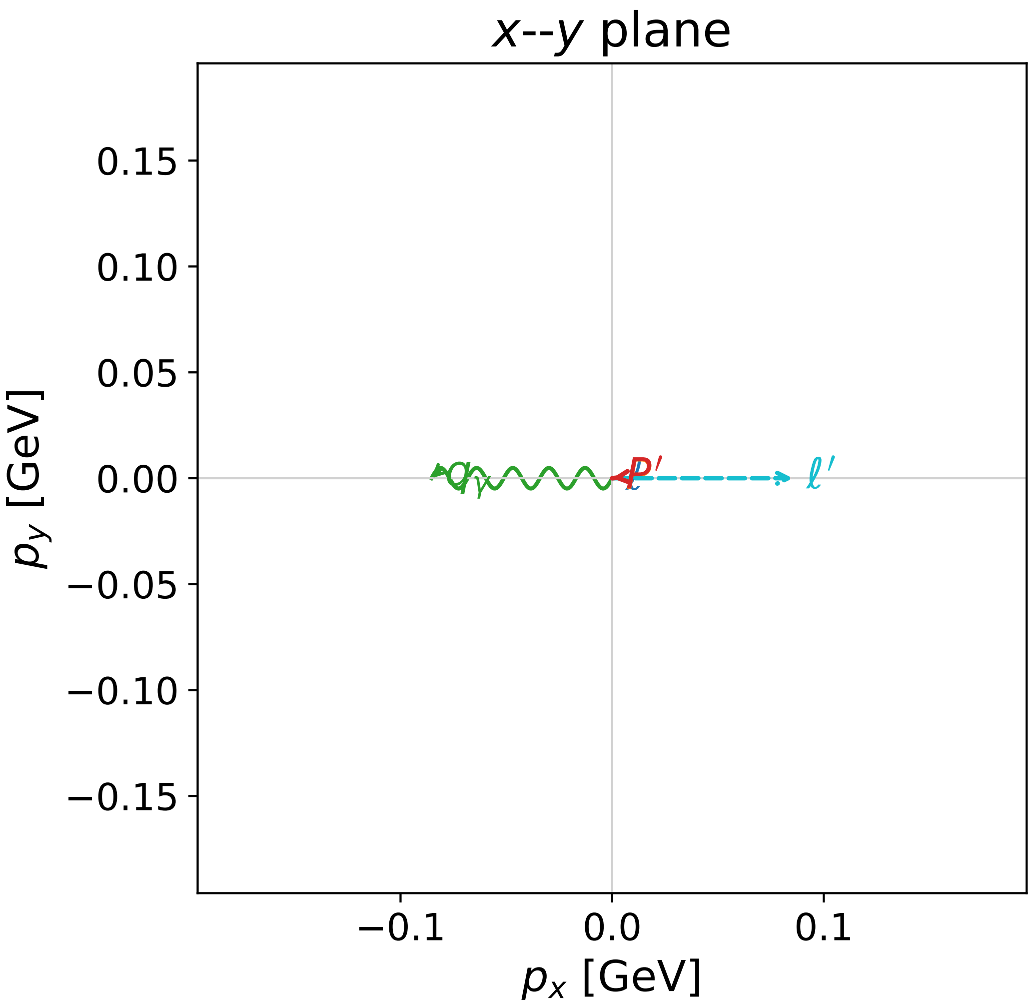
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

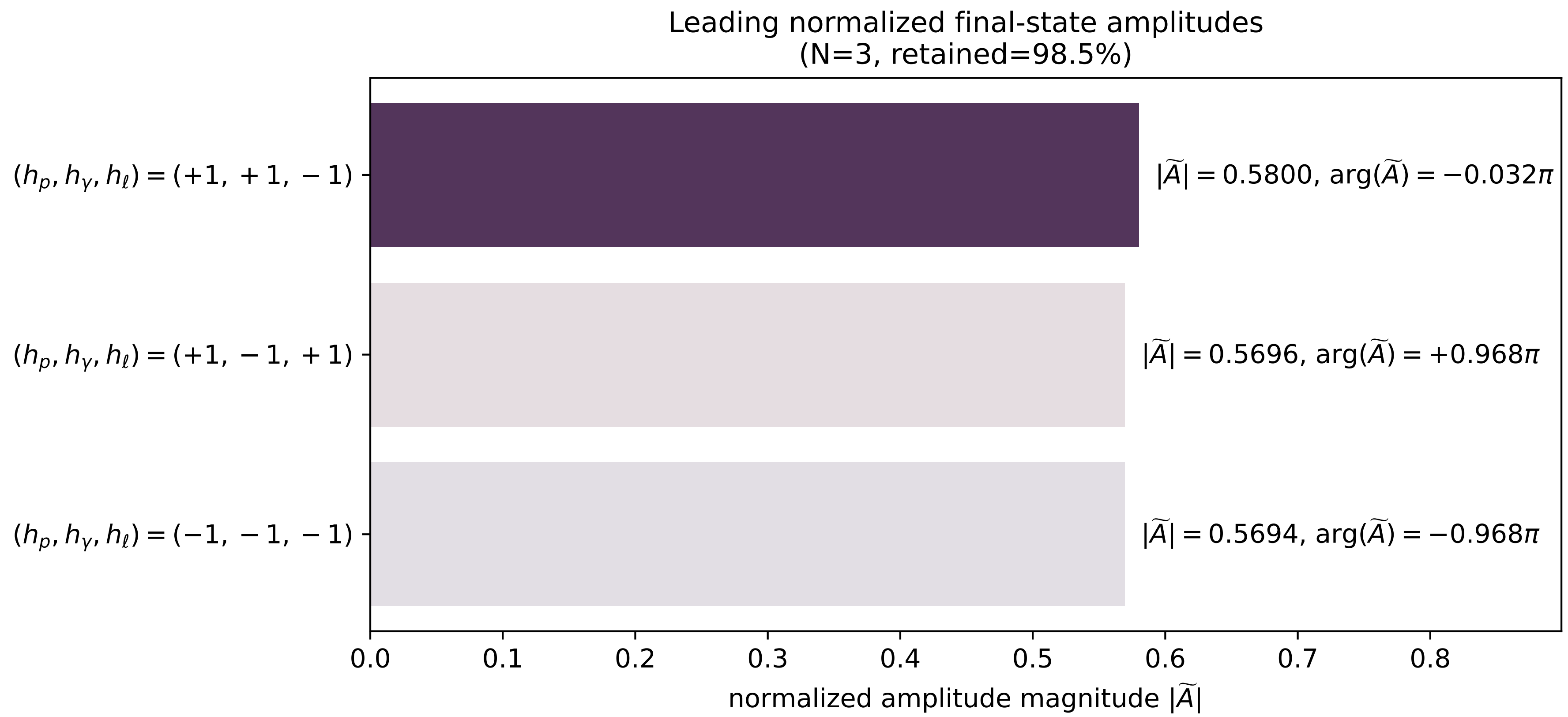



```
dw_mixing_angles_polarization_06_local_minimum_715
D_W=0.00790999
region: polarization_cluster_06_local_minimum_715
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0715
lepton mixing angle: alpha_e=2.3328713 rad
incoming lepton: -0.690424|+> +0.723405|->
proton mixing angle: alpha_p=1.763399 rad
incoming proton: -0.191414|+> +0.981509|->
```

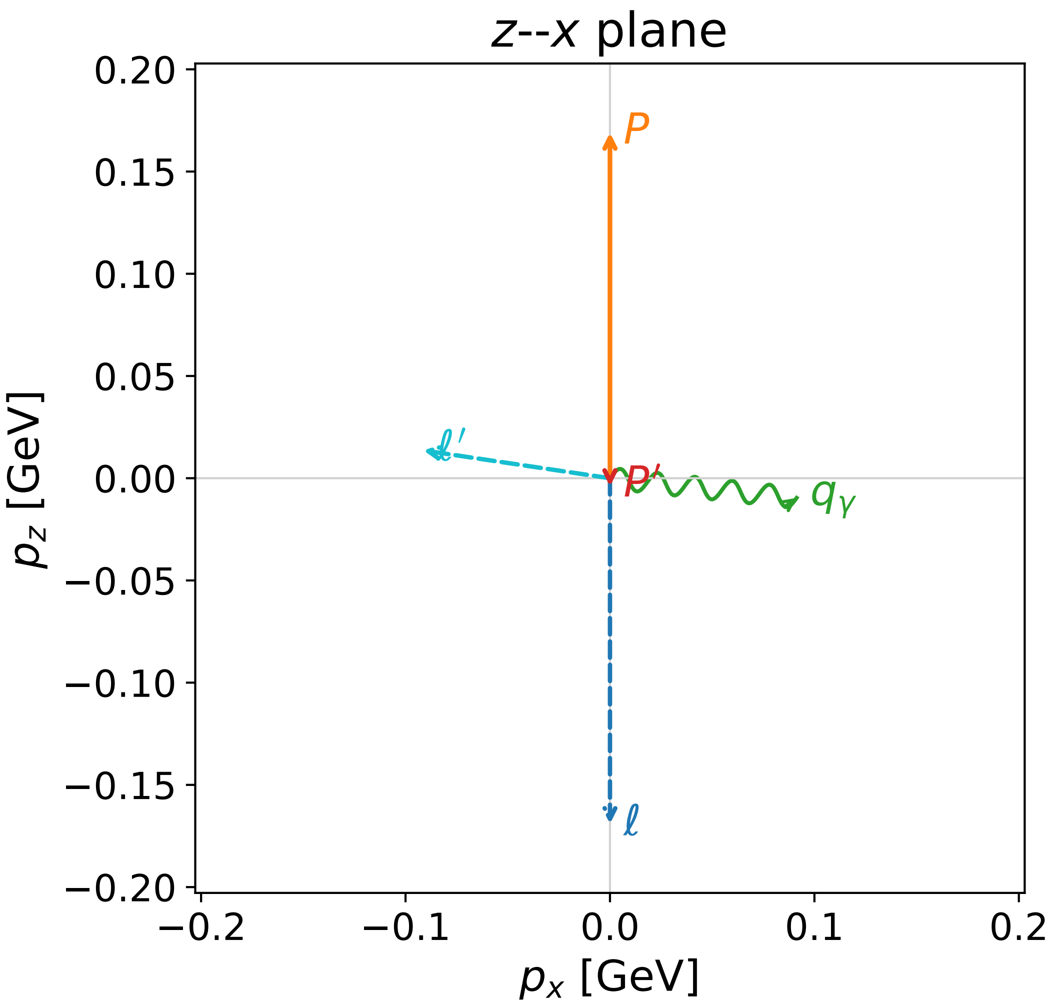
s=1.24396, $\sqrt{s}$=1.11533, $|\vec{P}|$=0.163233, $|\vec{P}'|$=0.000348371
$\theta_{p'}$=3.14148, θ_γ=1.8417, $\phi_{p'}$=3.24294, ϕ_γ=3.14161
E_γ=0.0886176, $\theta_{\ell'}$=1.29611, $\phi_{\ell'}$=1.5997e-05
Q^2=0.0368169, x_B=0.181322, t=-0.02656, W^2=1.04607, y=0.55764

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1632 p_x= 0.0000 p_y= 0.0000 p_z= -0.1632
P E= 0.9521 p_x= 0.0000 p_y= 0.0000 p_z= 0.1632
ℓ' E= 0.0887 p_x= 0.0854 p_y= 0.0000 p_z= 0.0241
P' E= 0.9380 p_x= -0.0000 p_y= -0.0000 p_z= -0.0003
q_γ E= 0.0886 p_x= -0.0854 p_y= -0.0000 p_z= -0.0237





coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

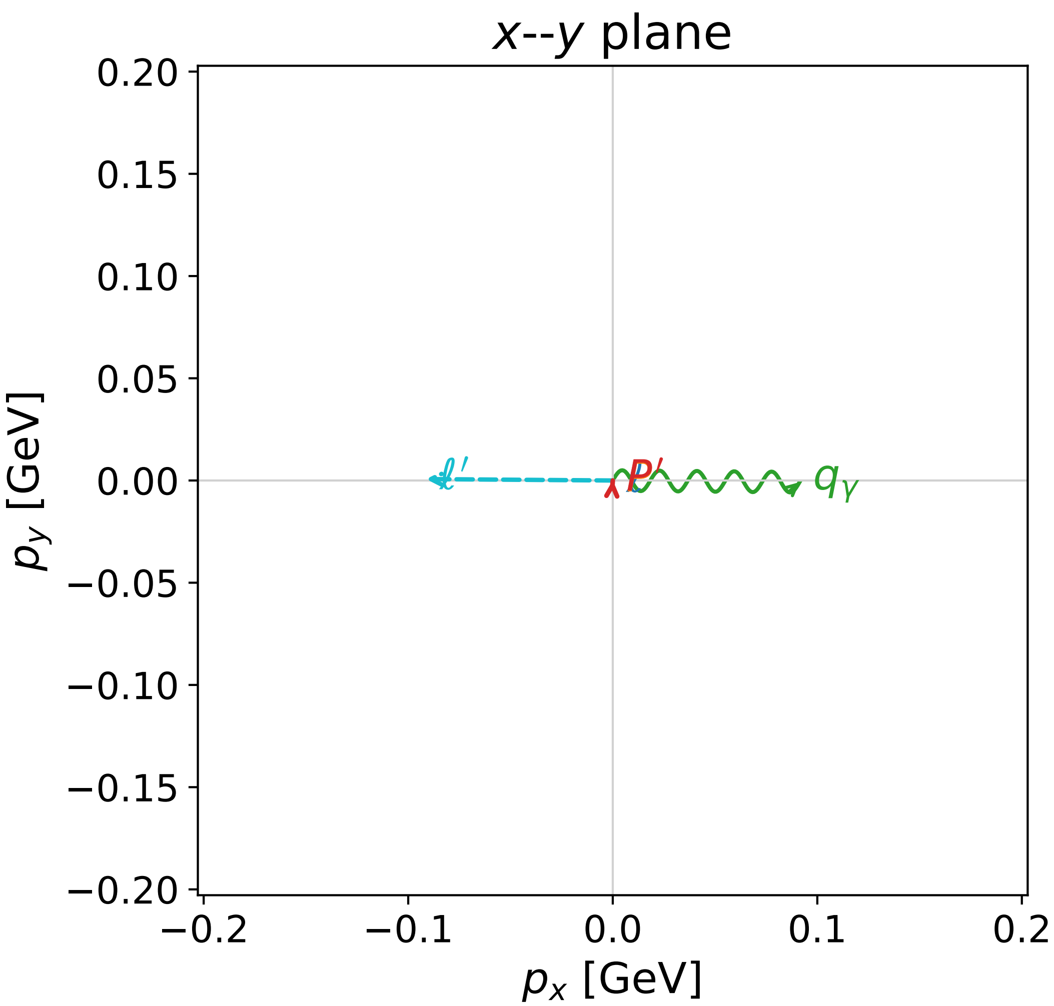


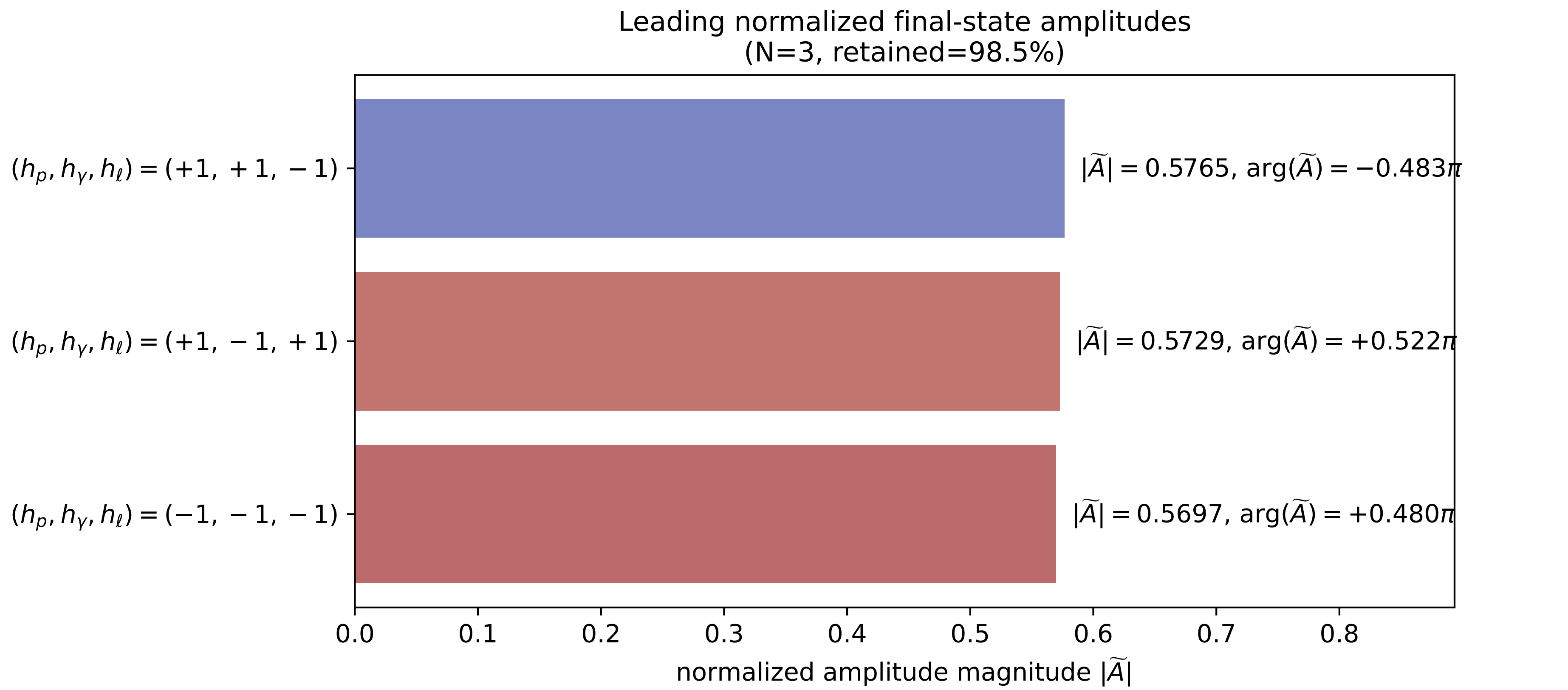

```
dw_mixing_angles_polarization_06_local_minimum_759
D_W=0.00935482
region: polarization_cluster_06_local_minimum_759
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0759
lepton mixing angle: alpha_e=2.3545656 rad
incoming lepton: -0.705954|+> +0.708258|->
proton mixing angle: alpha_p=1.3737025 rad
incoming proton: 0.19582|+> +0.98064|->
```

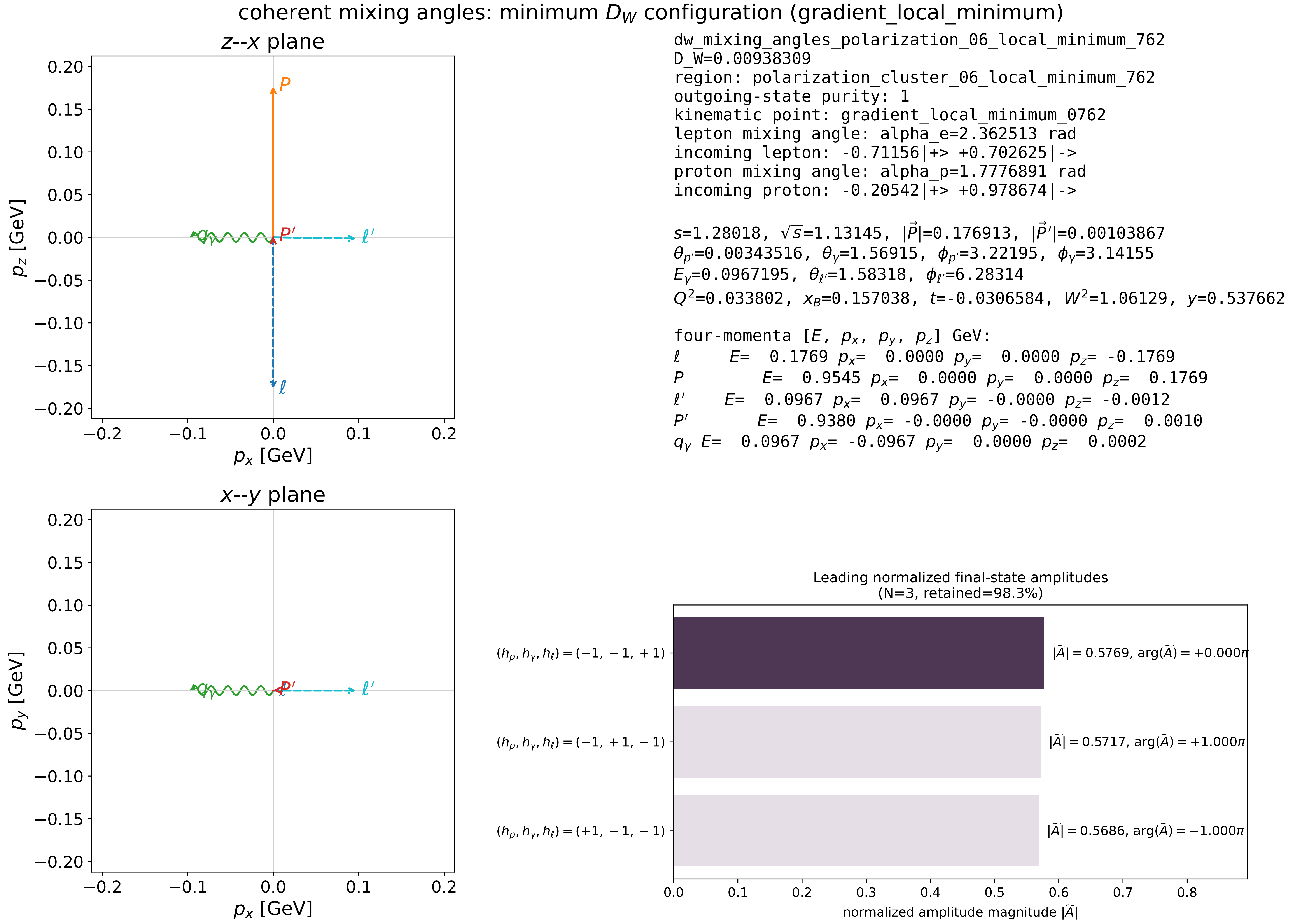
s=1.25922, $\sqrt{s}$=1.12215, $|\vec{P}|$=0.169039, $|\vec{P}'|$=0.00397886
$\theta_{p'}$=3.10698, θ_γ=1.67588, $\phi_{p'}$=1.50837, ϕ_γ=6.27405
E_γ=0.091815, $\theta_{\ell'}$=1.42287, $\phi_{\ell'}$=3.13397
Q^2=0.0358132, x_B=0.172194, t=-0.0297062, W^2=1.05201, y=0.548223

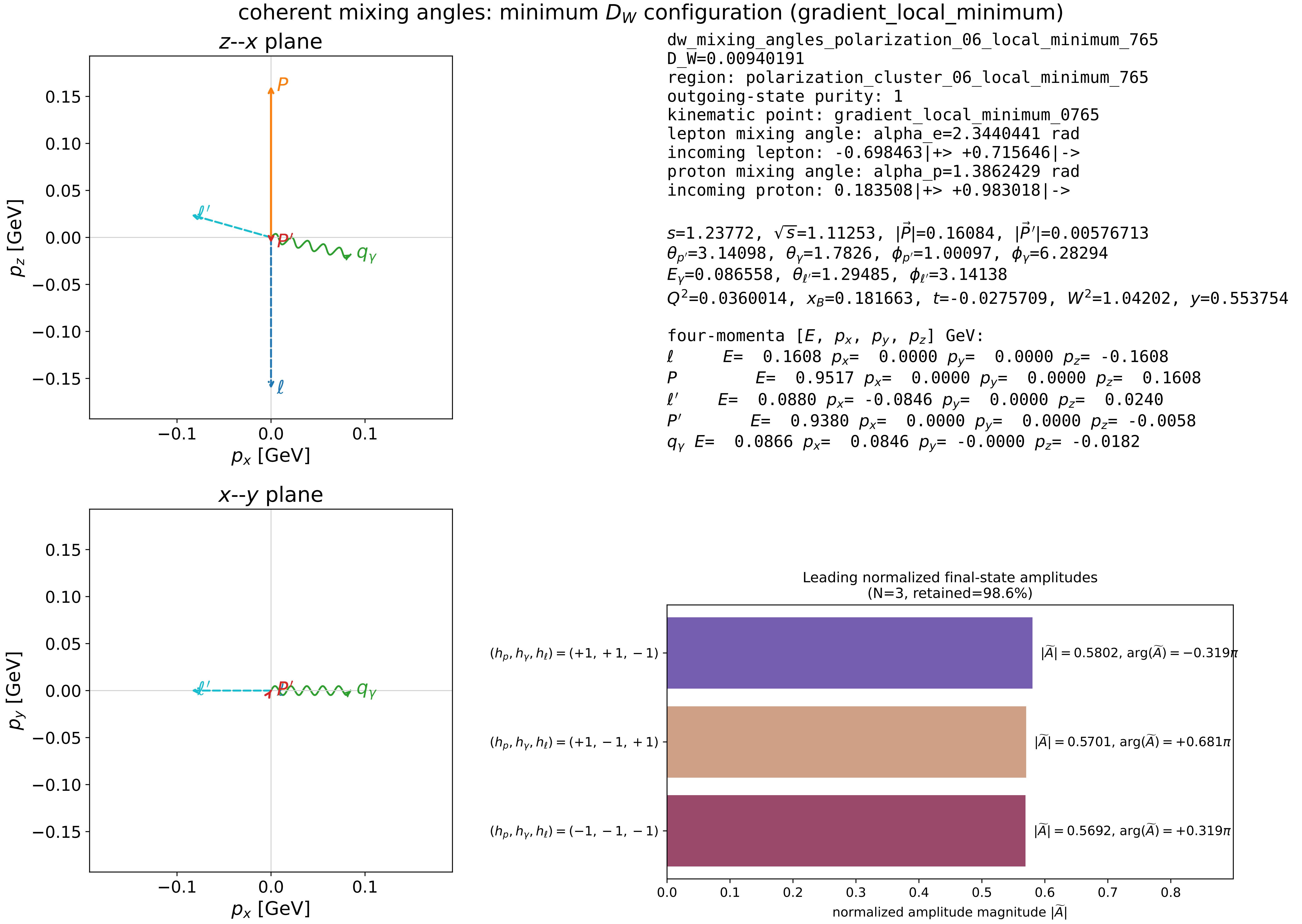
four-momenta [E, p_x, p_y, p_z] GeV:

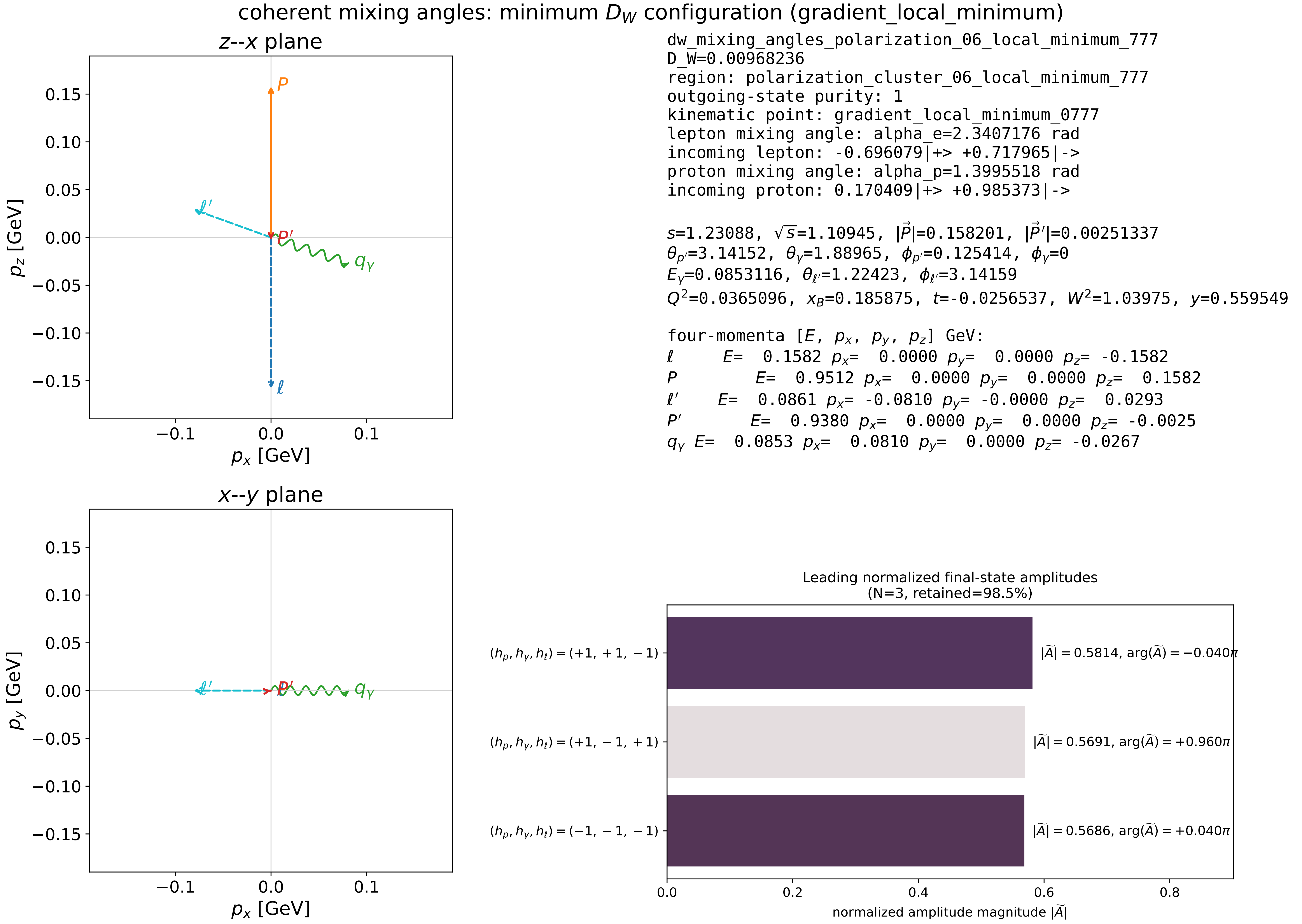
ℓ	E= 0.1690	p_x= 0.0000	p_y= 0.0000	p_z= -0.1690
P	E= 0.9531	p_x= 0.0000	p_y= 0.0000	p_z= 0.1690
ℓ'	E= 0.0923	p_x= -0.0913	p_y= 0.0007	p_z= 0.0136
P'	E= 0.9380	p_x= 0.0000	p_y= 0.0001	p_z= -0.0040
q_γ	E= 0.0918	p_x= 0.0913	p_y= -0.0008	p_z= -0.0096

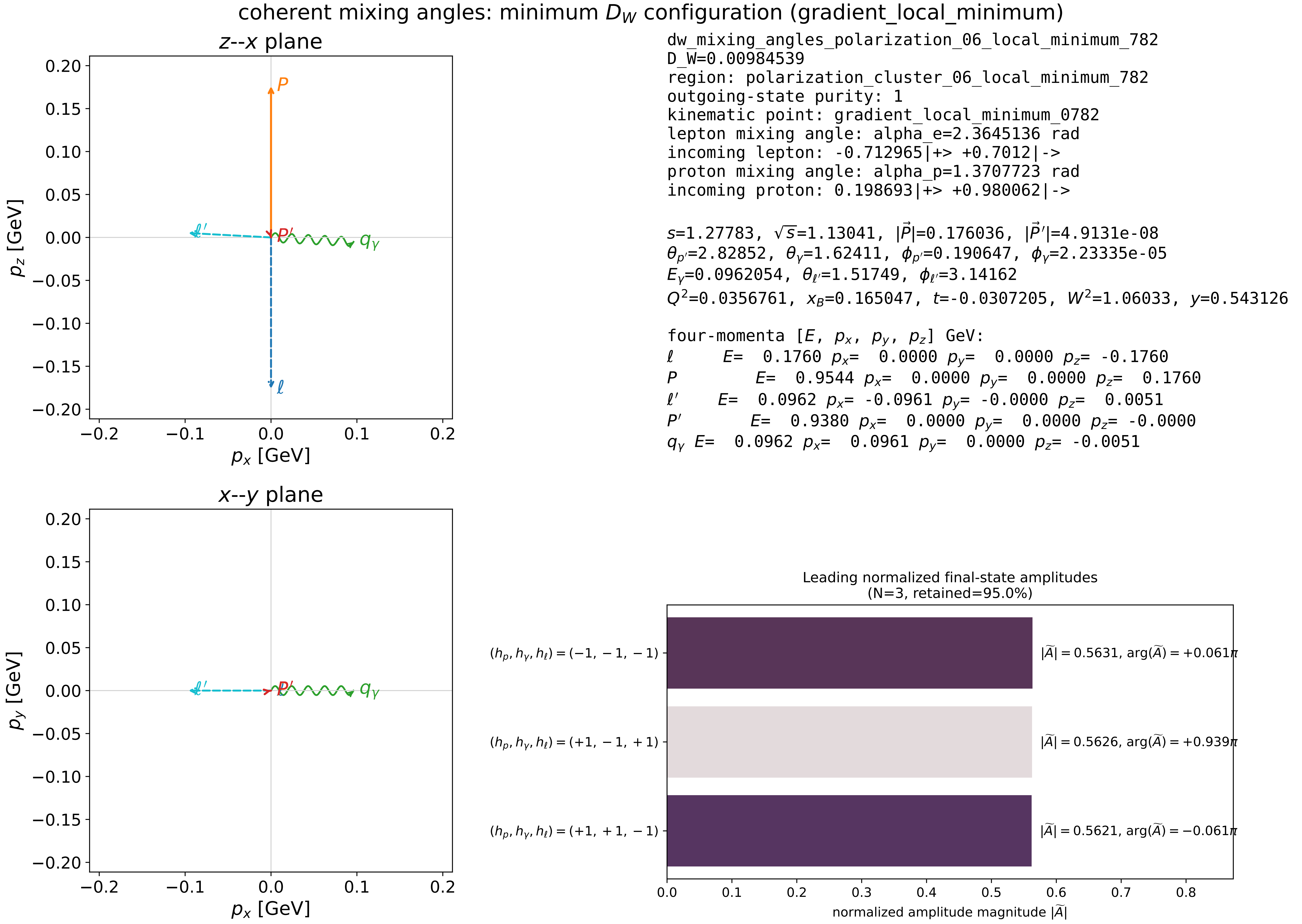





coherent mixing angles: minimum D_W configuration (gradient_local_minimum)
z--x plane
p_z [GeV]
p_x [GeV]
P
P'
ℓ'
q_γ
ℓ
x--y plane
p_y [GeV]
dw_mixing_angles_polarization_06_local_minimum_762
D_W=0.00938309
region: polarization_cluster_06_local_minimum_762
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0762
lepton mixing angle: alpha_e=2.362513 rad
incoming lepton: -0.71156|+> +0.702625|->
proton mixing angle: alpha_p=1.7776891 rad
incoming proton: -0.20542|+> +0.978674|->
s=1.28018, $\sqrt{s}$=1.13145, $|\vec{P}|$=0.176913, $|\vec{P}'|$=0.00103867
$\theta_{p'}$=0.00343516, θ_γ=1.56915, $\phi_{p'}$=3.22195, ϕ_γ=3.14155
E_γ=0.0967195, $\theta_{\ell'}$=1.58318, $\phi_{\ell'}$=6.28314
Q^2=0.033802, x_B=0.157038, t=-0.0306584, W^2=1.06129, y=0.537662
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1769 p_x= 0.0000 p_y= 0.0000 p_z= -0.1769
P E= 0.9545 p_x= 0.0000 p_y= 0.0000 p_z= 0.1769
ℓ' E= 0.0967 p_x= 0.0967 p_y= -0.0000 p_z= -0.0012
P' E= 0.9380 p_x= -0.0000 p_y= -0.0000 p_z= 0.0010
q_γ E= 0.0967 p_x= -0.0967 p_y= 0.0000 p_z= 0.0002
Leading normalized final-state amplitudes
(N=3, retained=98.3%)
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$
$|\widetilde{A}| = 0.5769$, $\arg(\widetilde{A}) = +0.000\pi$
$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$
$|\widetilde{A}| = 0.5717$, $\arg(\widetilde{A}) = +1.000\pi$
$(h_p, h_\gamma, h_\ell) = (+1, -1, -1)$
$|\widetilde{A}| = 0.5686$, $\arg(\widetilde{A}) = -1.000\pi$
normalized amplitude magnitude $|\widetilde{A}|$

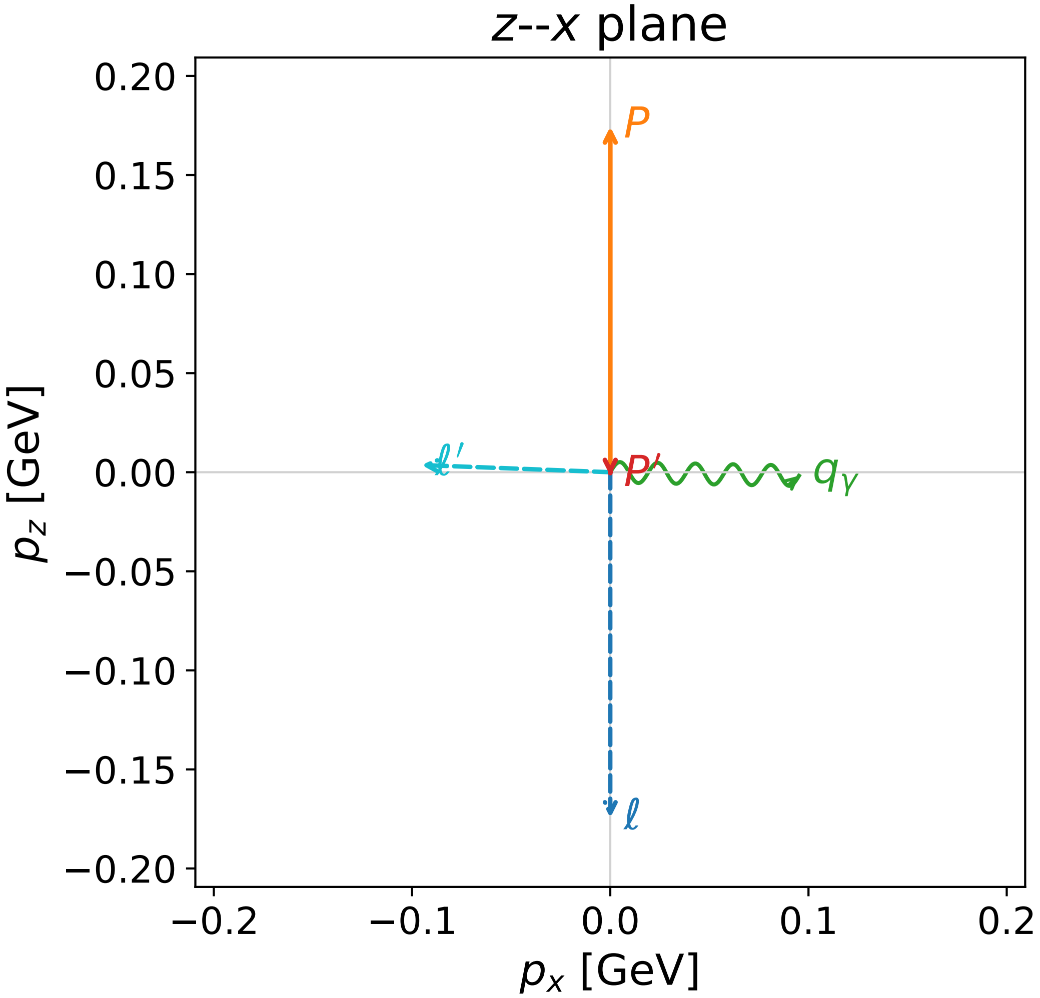

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)
z--x plane
P
ℓ'
P'
q_γ
ℓ
p_z [GeV]
p_x [GeV]
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
−0.1
0.0
0.1
x--y plane
p_y [GeV]
dw_mixing_angles_polarization_06_local_minimum_765
D_W=0.00940191
region: polarization_cluster_06_local_minimum_765
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0765
lepton mixing angle: alpha_e=2.3440441 rad
incoming lepton: -0.698463|+> +0.715646|->
proton mixing angle: alpha_p=1.3862429 rad
incoming proton: 0.183508|+> +0.983018|->
s=1.23772, $\sqrt{s}$=1.11253, $|\vec{P}|$=0.16084, $|\vec{P}'|$=0.00576713
$\theta_{p'}$=3.14098, θ_γ=1.7826, $\phi_{p'}$=1.00097, ϕ_γ=6.28294
E_γ=0.086558, $\theta_{\ell'}$=1.29485, $\phi_{\ell'}$=3.14138
Q^2=0.0360014, x_B=0.181663, t=-0.0275709, W^2=1.04202, y=0.553754
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1608 p_x= 0.0000 p_y= 0.0000 p_z= -0.1608
P E= 0.9517 p_x= 0.0000 p_y= 0.0000 p_z= 0.1608
ℓ' E= 0.0880 p_x= -0.0846 p_y= 0.0000 p_z= 0.0240
P' E= 0.9380 p_x= 0.0000 p_y= 0.0000 p_z= -0.0058
q_γ E= 0.0866 p_x= 0.0846 p_y= -0.0000 p_z= -0.0182
Leading normalized final-state amplitudes
(N=3, retained=98.6%)
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$
$|\widetilde{A}| = 0.5802$, $\arg(\widetilde{A}) = -0.319\pi$
$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$
$|\widetilde{A}| = 0.5701$, $\arg(\widetilde{A}) = +0.681\pi$
$(h_p, h_\gamma, h_\ell) = (-1, -1, -1)$
$|\widetilde{A}| = 0.5692$, $\arg(\widetilde{A}) = +0.319\pi$
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
normalized amplitude magnitude $|\widetilde{A}|$


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)
z--x plane
p_z [GeV]
p_x [GeV]
P
ℓ'
P'
q_γ
ℓ
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
−0.1
0.0
0.1
dw_mixing_angles_polarization_06_local_minimum_777
D_W=0.00968236
region: polarization_cluster_06_local_minimum_777
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0777
lepton mixing angle: alpha_e=2.3407176 rad
incoming lepton: -0.696079|+> +0.717965|->
proton mixing angle: alpha_p=1.3995518 rad
incoming proton: 0.170409|+> +0.985373|->
s=1.23088, $\sqrt{s}$=1.10945, $|\vec{P}|$=0.158201, $|\vec{P}'|$=0.00251337
$\theta_{p'}$=3.14152, θ_γ=1.88965, $\phi_{p'}$=0.125414, ϕ_γ=0
E_γ=0.0853116, $\theta_{\ell'}$=1.22423, $\phi_{\ell'}$=3.14159
Q^2=0.0365096, x_B=0.185875, t=-0.0256537, W^2=1.03975, y=0.559549
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1582 p_x= 0.0000 p_y= 0.0000 p_z= -0.1582
P E= 0.9512 p_x= 0.0000 p_y= 0.0000 p_z= 0.1582
ℓ' E= 0.0861 p_x= -0.0810 p_y= -0.0000 p_z= 0.0293
P' E= 0.9380 p_x= 0.0000 p_y= 0.0000 p_z= -0.0025
q_γ E= 0.0853 p_x= 0.0810 p_y= 0.0000 p_z= -0.0267
x--y plane
p_y [GeV]
p_x [GeV]
ℓ'
P'
q_γ
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
−0.1
0.0
0.1
Leading normalized final-state amplitudes
(N=3, retained=98.5%)
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$
$|\widetilde{A}| = 0.5814$, $\arg(\widetilde{A}) = -0.040\pi$
$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$
$|\widetilde{A}| = 0.5691$, $\arg(\widetilde{A}) = +0.960\pi$
$(h_p, h_\gamma, h_\ell) = (-1, -1, -1)$
$|\widetilde{A}| = 0.5686$, $\arg(\widetilde{A}) = +0.040\pi$
0.0
0.2
0.4
0.6
0.8
normalized amplitude magnitude $|\widetilde{A}|$



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)
z--x plane
P
ℓ'
P'
q_γ
ℓ
p_z [GeV]
p_x [GeV]
dw_mixing_angles_polarization_06_local_minimum_782
D_W=0.00984539
region: polarization_cluster_06_local_minimum_782
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0782
lepton mixing angle: alpha_e=2.3645136 rad
incoming lepton: -0.712965|+> +0.7012|->
proton mixing angle: alpha_p=1.3707723 rad
incoming proton: 0.198693|+> +0.980062|->
s=1.27783, $\sqrt{s}$=1.13041, $|\vec{P}|$=0.176036, $|\vec{P}'|$=4.9131e-08
$\theta_{p'}$=2.82852, θ_γ=1.62411, $\phi_{p'}$=0.190647, ϕ_γ=2.23335e-05
E_γ=0.0962054, $\theta_{\ell'}$=1.51749, $\phi_{\ell'}$=3.14162
Q^2=0.0356761, x_B=0.165047, t=-0.0307205, W^2=1.06033, y=0.543126
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1760 p_x= 0.0000 p_y= 0.0000 p_z= -0.1760
P E= 0.9544 p_x= 0.0000 p_y= 0.0000 p_z= 0.1760
ℓ' E= 0.0962 p_x= -0.0961 p_y= -0.0000 p_z= 0.0051
P' E= 0.9380 p_x= 0.0000 p_y= 0.0000 p_z= -0.0000
q_γ E= 0.0962 p_x= 0.0961 p_y= 0.0000 p_z= -0.0051
x--y plane
p_y [GeV]
p_x [GeV]
Leading normalized final-state amplitudes
(N=3, retained=95.0%)
$(h_p, h_\gamma, h_\ell) = (-1, -1, -1)$
$|\widetilde{A}| = 0.5631$, $\arg(\widetilde{A}) = +0.061\pi$
$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$
$|\widetilde{A}| = 0.5626$, $\arg(\widetilde{A}) = +0.939\pi$
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$
$|\widetilde{A}| = 0.5621$, $\arg(\widetilde{A}) = -0.061\pi$
normalized amplitude magnitude $|\widetilde{A}|$

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

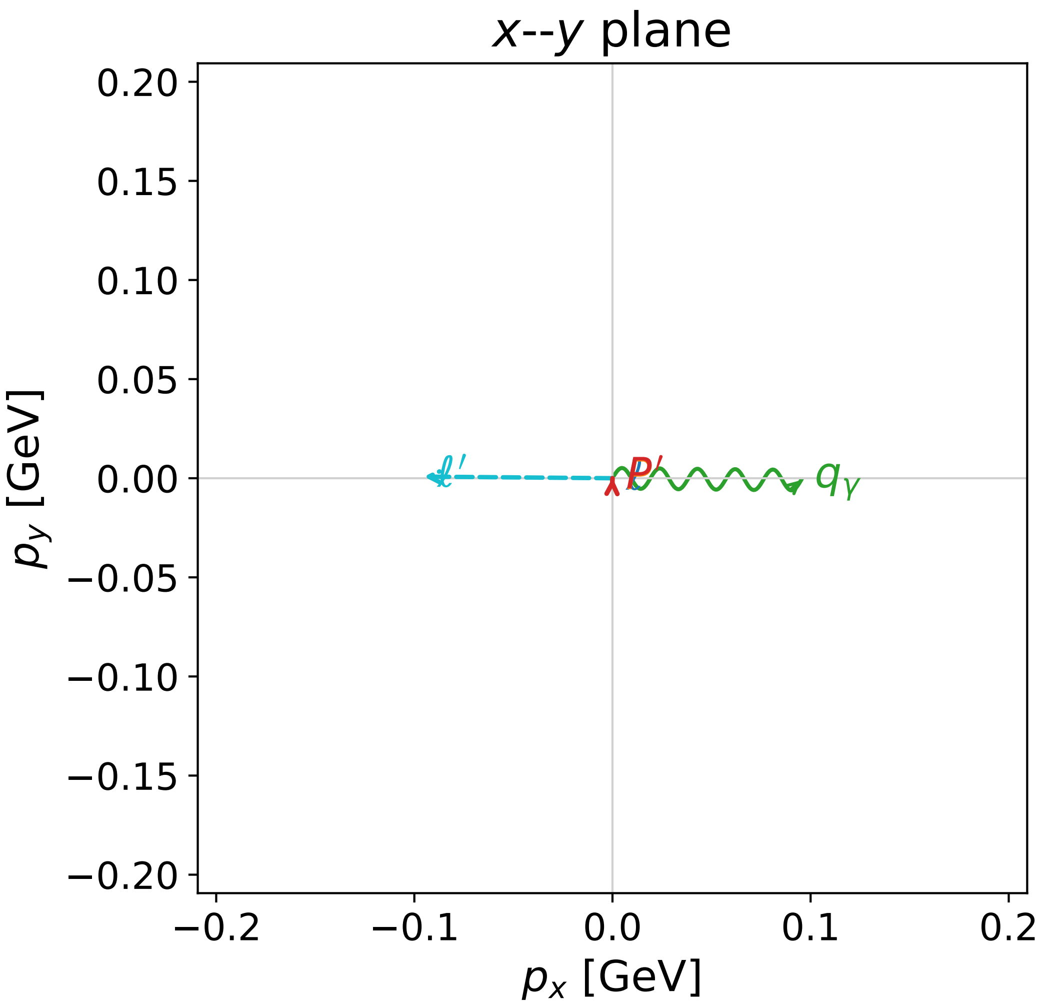


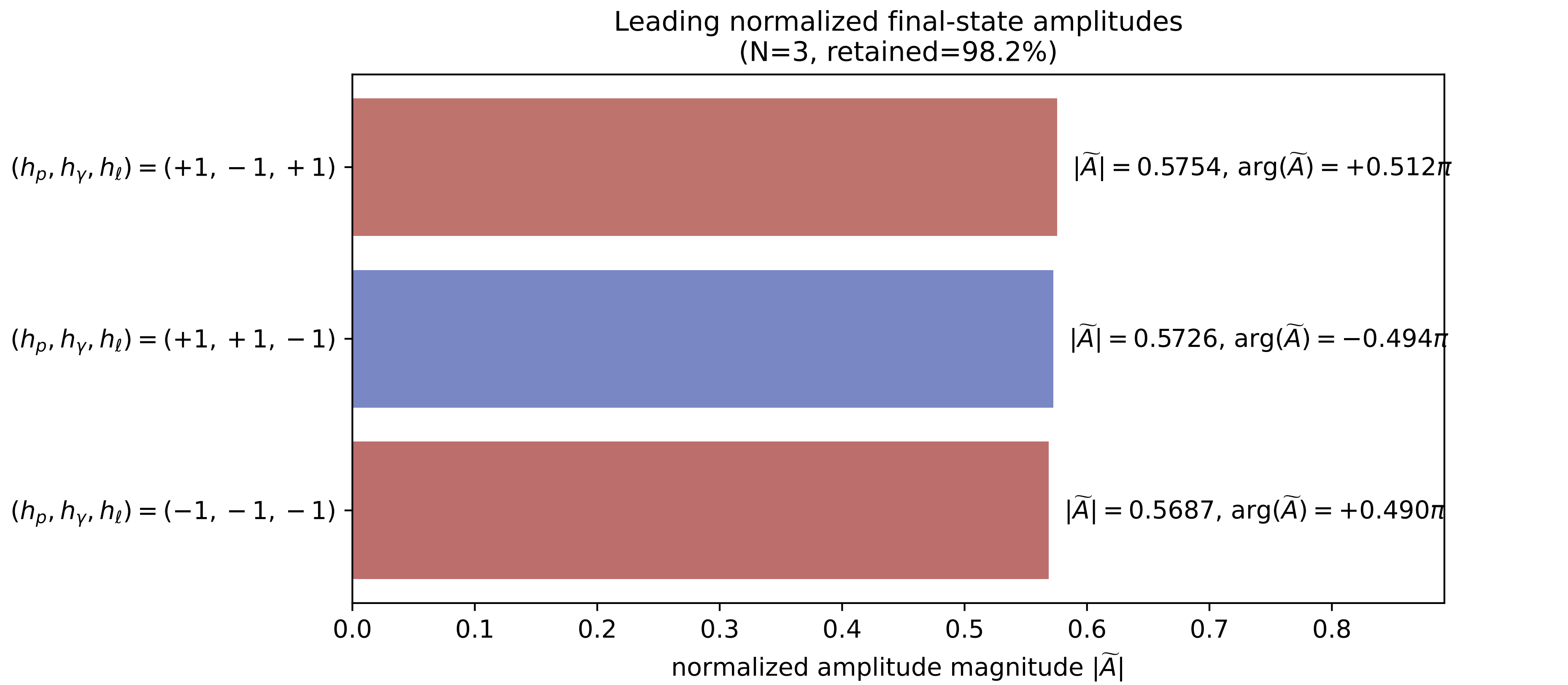

```
dw_mixing_angles_polarization_06_local_minimum_786
D_W=0.0099536
region: polarization_cluster_06_local_minimum_786
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0786
lepton mixing angle: alpha_e=2.3653262 rad
incoming lepton: -0.713534|+> +0.70062|->
proton mixing angle: alpha_p=1.3705843 rad
incoming proton: 0.198877|+> +0.980024|->
```

s=1.27353, $\sqrt{s}$=1.12851, $|\vec{P}|$=0.174427, $|\vec{P}'|$=0.00184551
$\theta_{p'}$=3.05371, θ_γ=1.58959, $\phi_{p'}$=1.53975, ϕ_γ=6.27289
E_γ=0.0952242, $\theta_{\ell'}$=1.53271, $\phi_{\ell'}$=3.133
Q^2=0.0345044, x_B=0.161888, t=-0.0308108, W^2=1.05848, y=0.541393

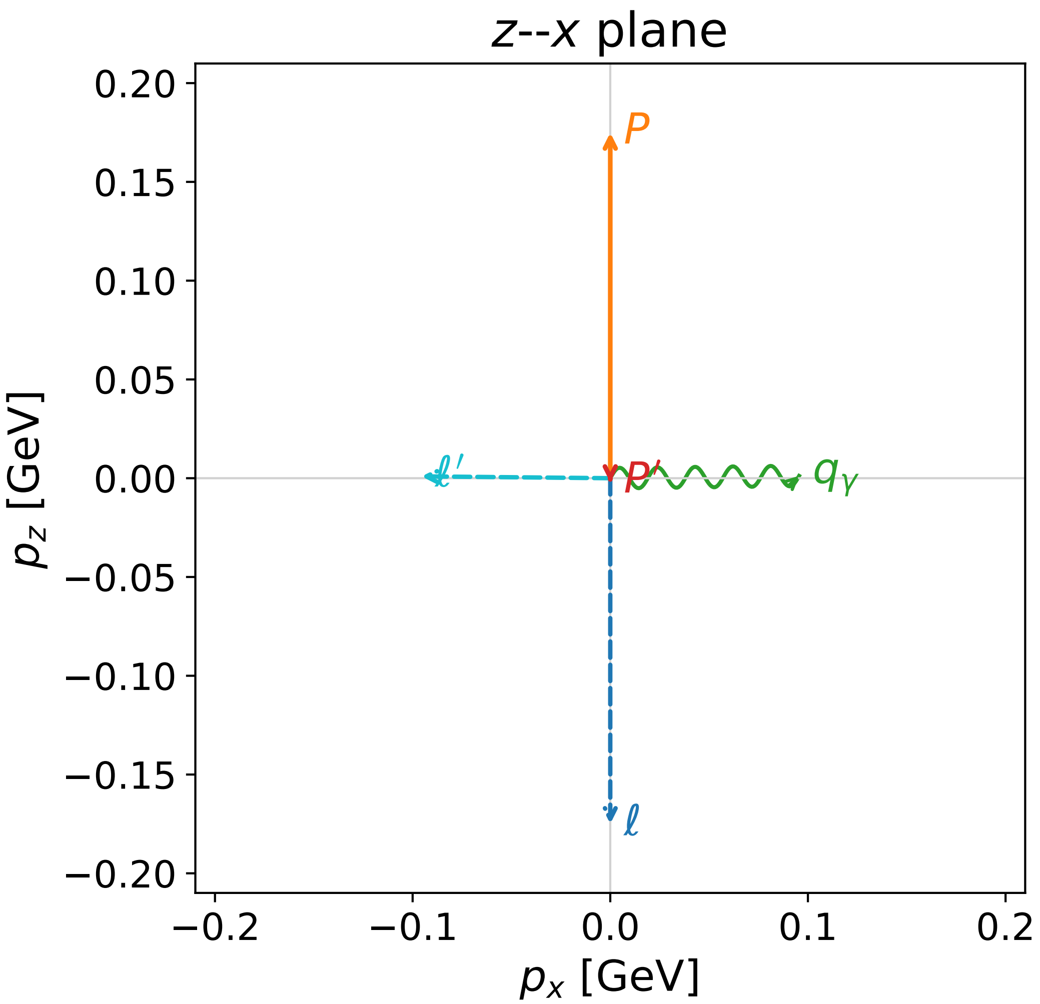
four-momenta [E, p_x, p_y, p_z] GeV:

ℓ	E= 0.1744	p_x= 0.0000	p_y= 0.0000	p_z= -0.1744
P	E= 0.9541	p_x= 0.0000	p_y= 0.0000	p_z= 0.1744
ℓ'	E= 0.0953	p_x= -0.0952	p_y= 0.0008	p_z= 0.0036
P'	E= 0.9380	p_x= 0.0000	p_y= 0.0002	p_z= -0.0018
q_γ	E= 0.0952	p_x= 0.0952	p_y= -0.0010	p_z= -0.0018





coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

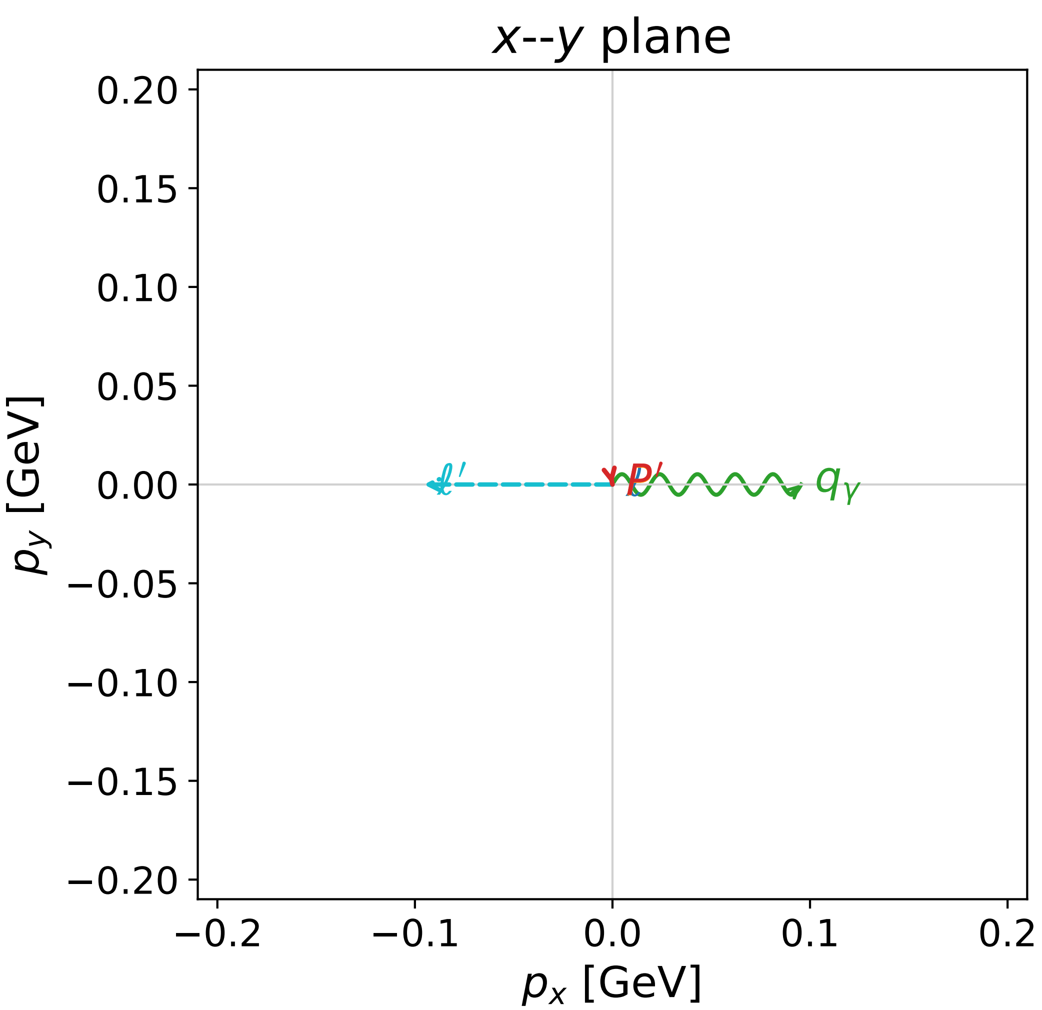


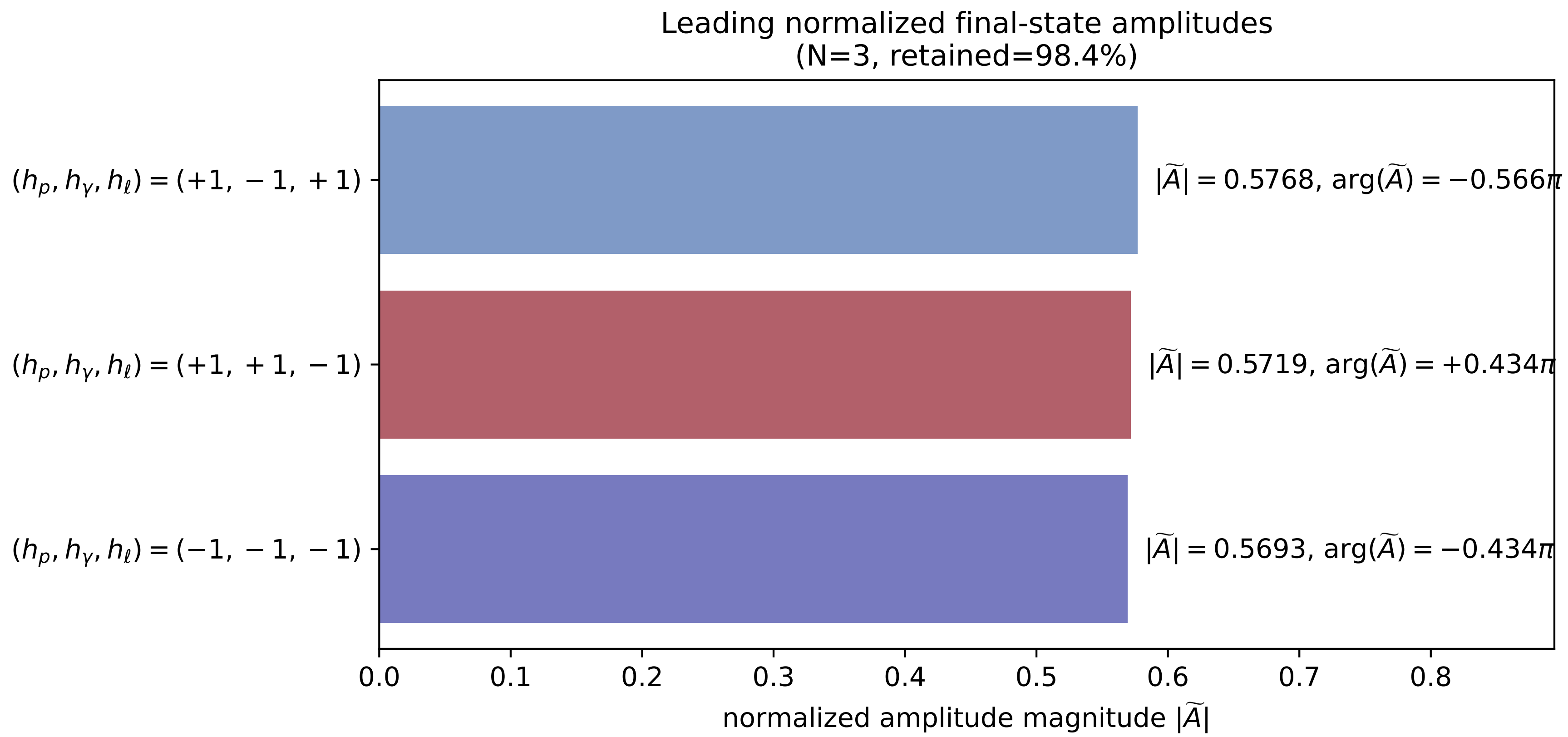

```
dw_mixing_angles_polarization_06_local_minimum_795
D_W=0.0102153
region: polarization_cluster_06_local_minimum_795
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0795
lepton mixing angle: alpha_e=2.3684745 rad
incoming lepton: -0.715737|+> +0.69837|->
proton mixing angle: alpha_p=1.3705715 rad
incoming proton: 0.19889|+> +0.980022|->
```

s=1.2749, $\sqrt{s}$=1.12911, $|\vec{P}|$=0.174939, $|\vec{P}'|$=0.00210552
$\theta_{p'}$=3.14096, θ_γ=1.55855, $\phi_{p'}$=4.91864, ϕ_γ=0.000108903
E_γ=0.095556, $\theta_{\ell'}$=1.56101, $\phi_{\ell'}$=3.14169
Q^2=0.0337594, x_B=0.158474, t=-0.0310832, W^2=1.05911, y=0.539239

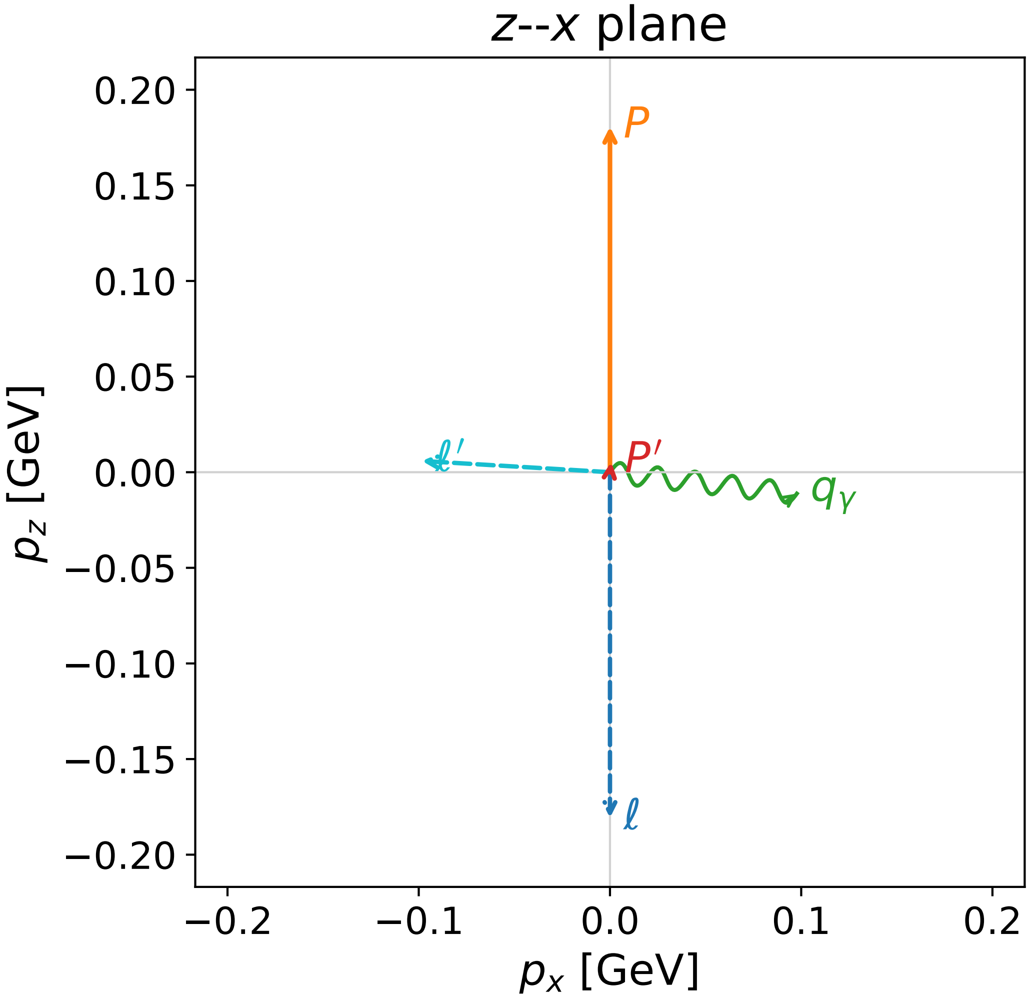
four-momenta [E, p_x, p_y, p_z] GeV:

ℓ	E= 0.1749	p_x= 0.0000	p_y= 0.0000	p_z= -0.1749
P	E= 0.9542	p_x= 0.0000	p_y= 0.0000	p_z= 0.1749
ℓ'	E= 0.0956	p_x= -0.0955	p_y= -0.0000	p_z= 0.0009
P'	E= 0.9380	p_x= 0.0000	p_y= -0.0000	p_z= -0.0021
q_γ	E= 0.0956	p_x= 0.0955	p_y= 0.0000	p_z= 0.0012





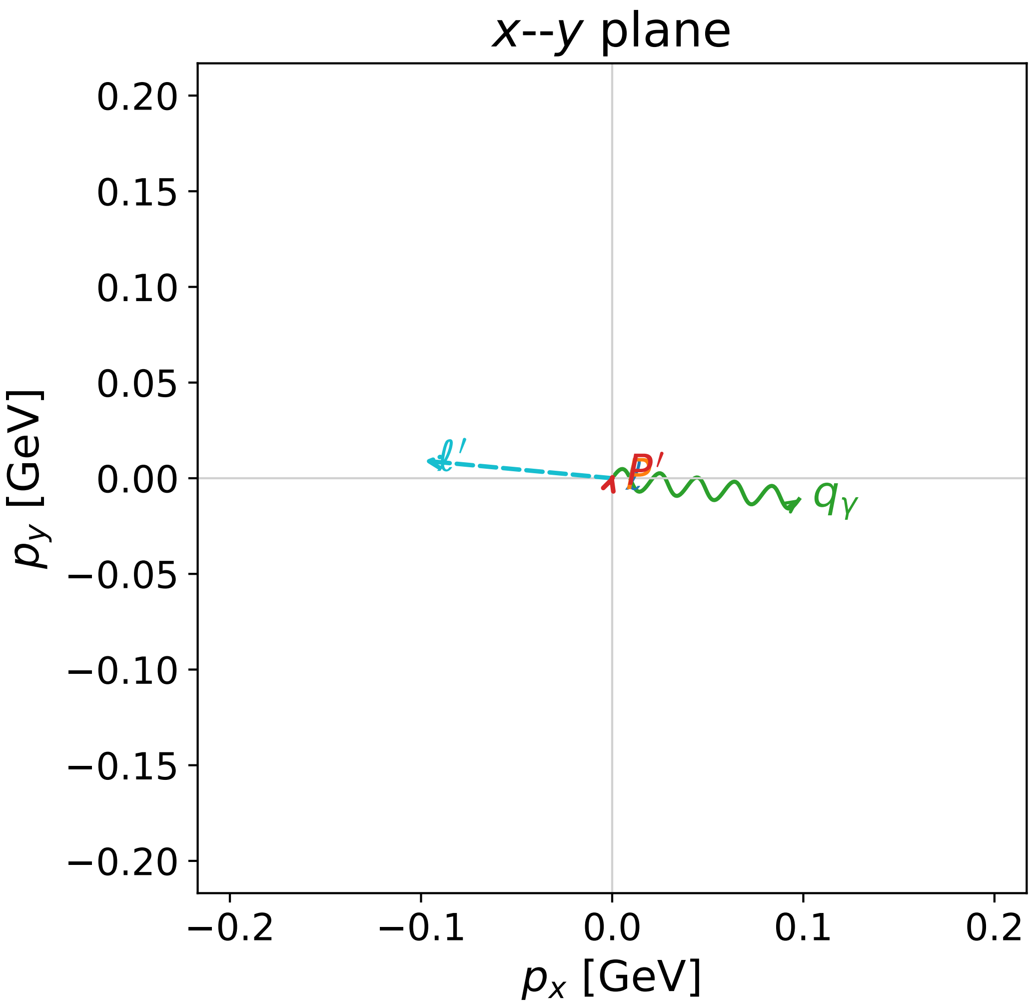
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

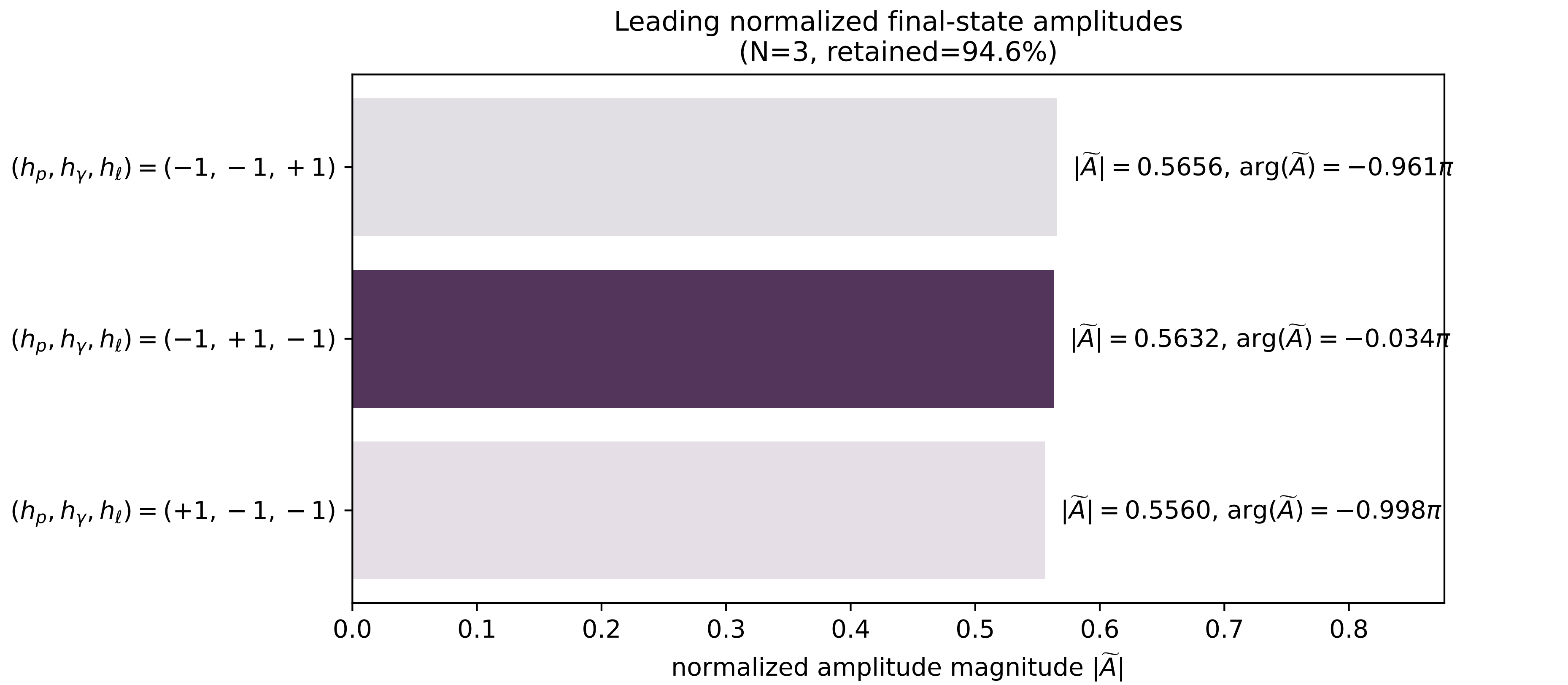


dw_mixing_angles_polarization_06_local_minimum_812
D_W=0.0107861
region: polarization_cluster_06_local_minimum_812
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0812
lepton mixing angle: alpha_e=2.3638088 rad
incoming lepton: -0.71247|+> +0.701702|->
proton mixing angle: alpha_p=1.3635838 rad
incoming proton: 0.205733|+> +0.978608|->

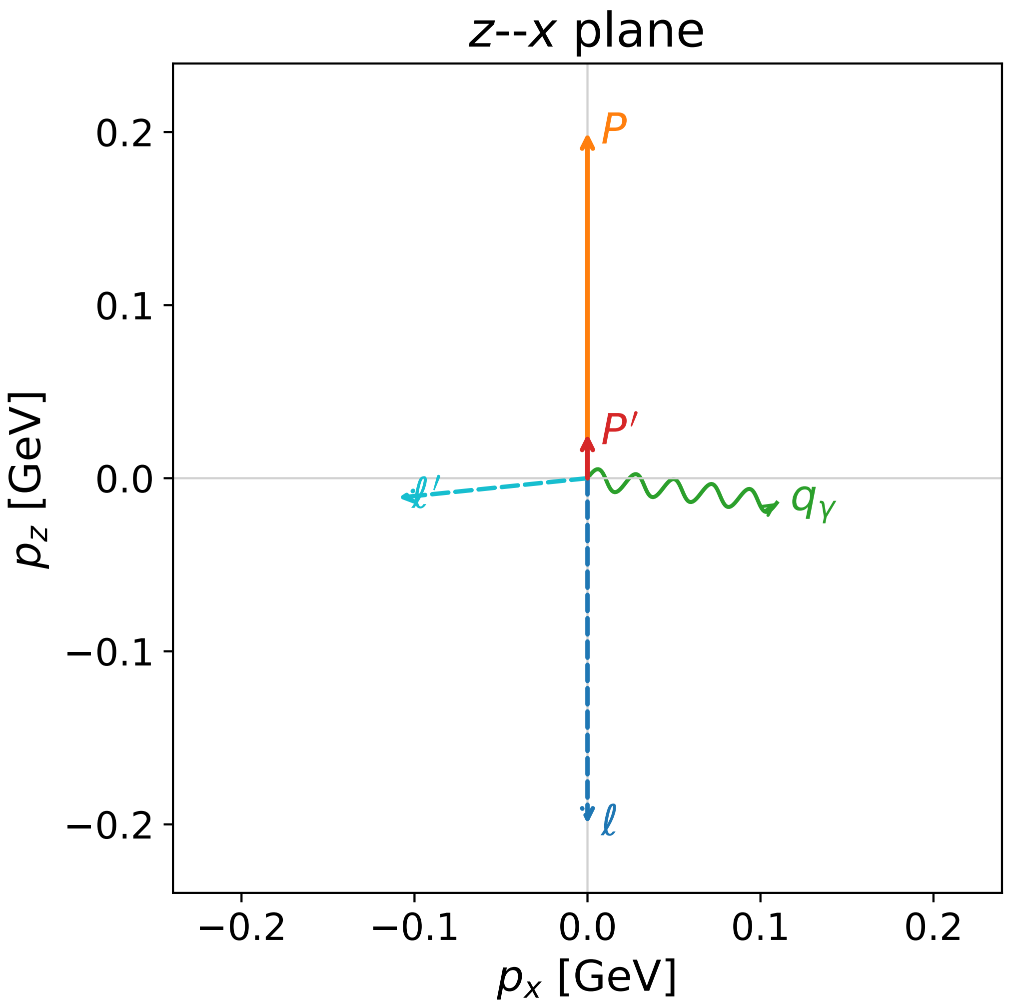
s=1.29042, $\sqrt{s}$=1.13597, $|\vec{P}|$=0.180716, $|\vec{P}'|$=0.00566642
$\theta_{p'}$=0.359605, θ_γ=1.68413, $\phi_{p'}$=1.24282, ϕ_γ=6.1708
E_γ=0.0989809, $\theta_{\ell'}$=1.51125, $\phi_{\ell'}$=3.04893
Q^2=0.0378986, x_B=0.169474, t=-0.0304764, W^2=1.06557, y=0.544662

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1807 p_x= 0.0000 p_y= 0.0000 p_z= -0.1807
P E= 0.9552 p_x= 0.0000 p_y= 0.0000 p_z= 0.1807
ℓ' E= 0.0990 p_x= -0.0984 p_y= 0.0091 p_z= 0.0059
P' E= 0.9380 p_x= 0.0006 p_y= 0.0019 p_z= 0.0053
q_γ E= 0.0990 p_x= 0.0977 p_y= -0.0110 p_z= -0.0112





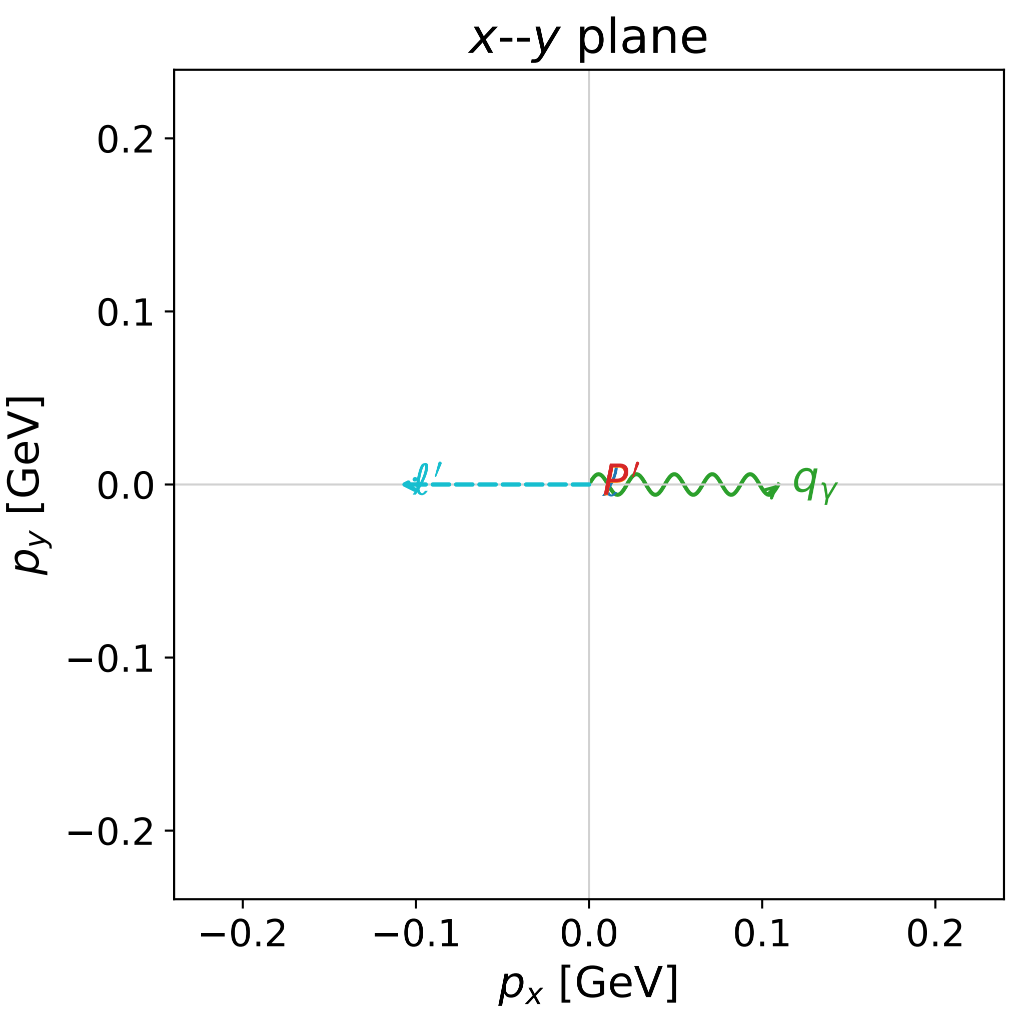
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

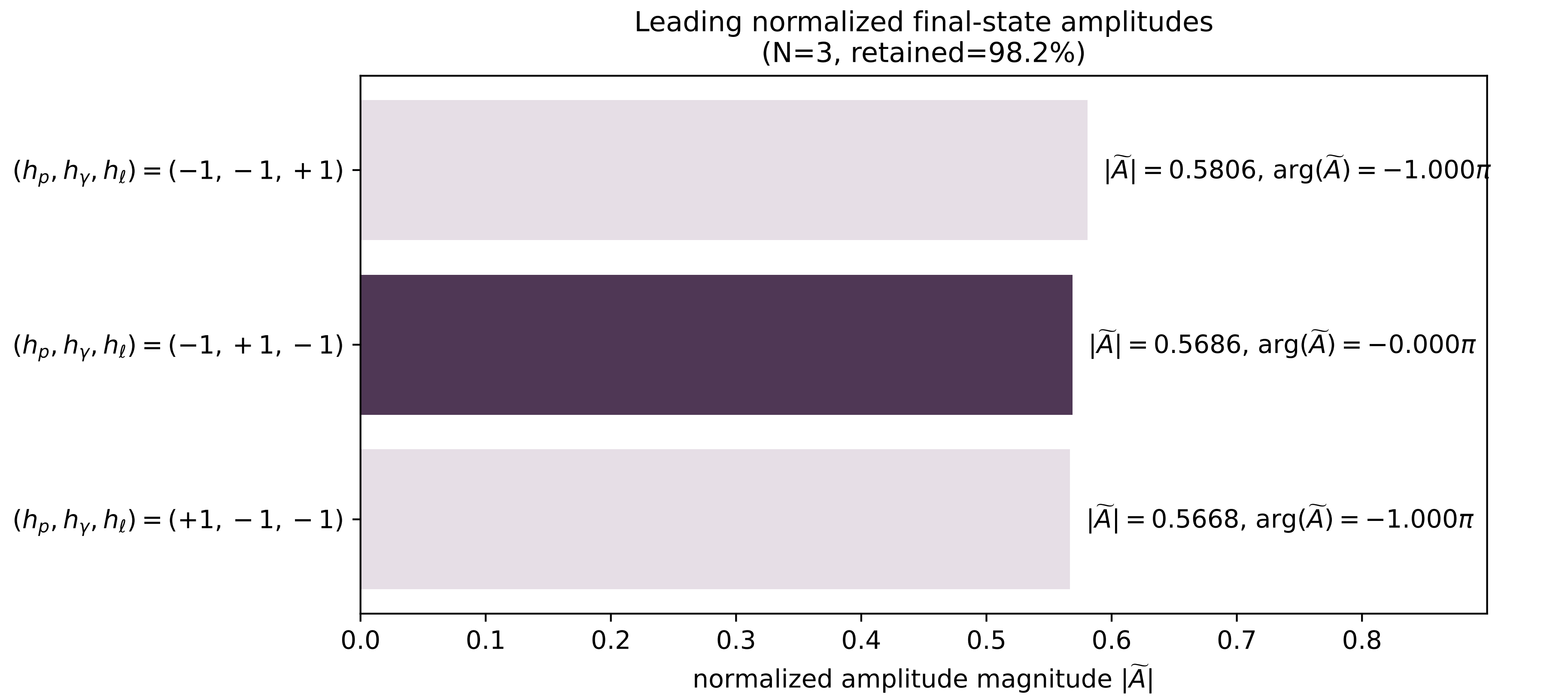


dw_mixing_angles_polarization_06_local_minimum_831
D_W=0.0114378
region: polarization_cluster_06_local_minimum_831
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0831
lepton mixing angle: alpha_e=2.3807655 rad
incoming lepton: -0.724266|+> +0.689521|->
proton mixing angle: alpha_p=1.3809873 rad
incoming proton: 0.188671|+> +0.98204|->

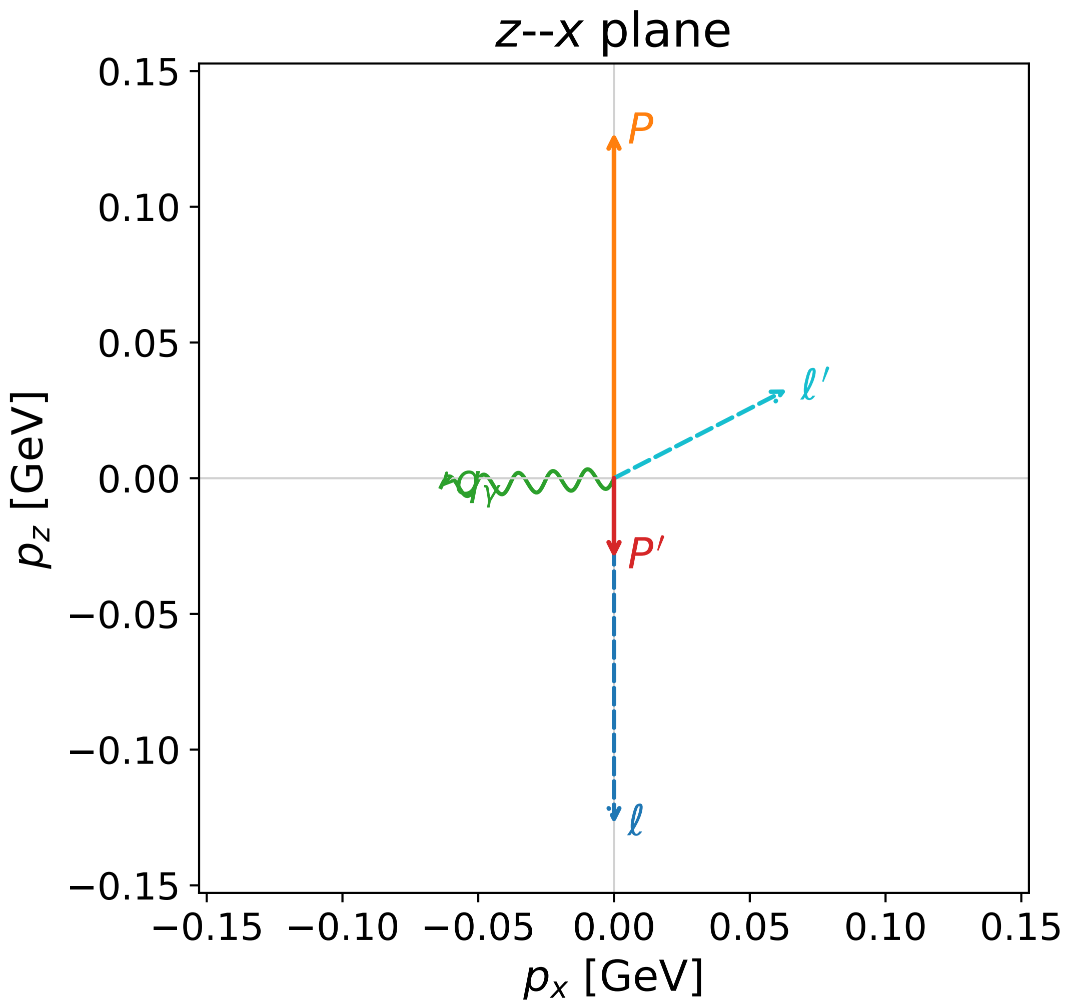
s=1.34258, $\sqrt{s}$=1.1587, $|\vec{P}|$=0.199679, $|\vec{P}'|$=0.0255397
$\theta_{p'}$=0, θ_γ=1.70042, $\phi_{p'}$=0.802418, ϕ_γ=6.28318
E_γ=0.110347, $\theta_{\ell'}$=1.67348, $\phi_{\ell'}$=3.14158
Q^2=0.0394268, x_B=0.159465, t=-0.0298971, W^2=1.08766, y=0.53431

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1997 p_x= 0.0000 p_y= 0.0000 p_z= -0.1997
P E= 0.9590 p_x= 0.0000 p_y= 0.0000 p_z= 0.1997
ℓ' E= 0.1100 p_x= -0.1094 p_y= 0.0000 p_z= -0.0113
P' E= 0.9383 p_x= 0.0000 p_y= 0.0000 p_z= 0.0255
q_γ E= 0.1103 p_x= 0.1094 p_y= -0.0000 p_z= -0.0143





coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

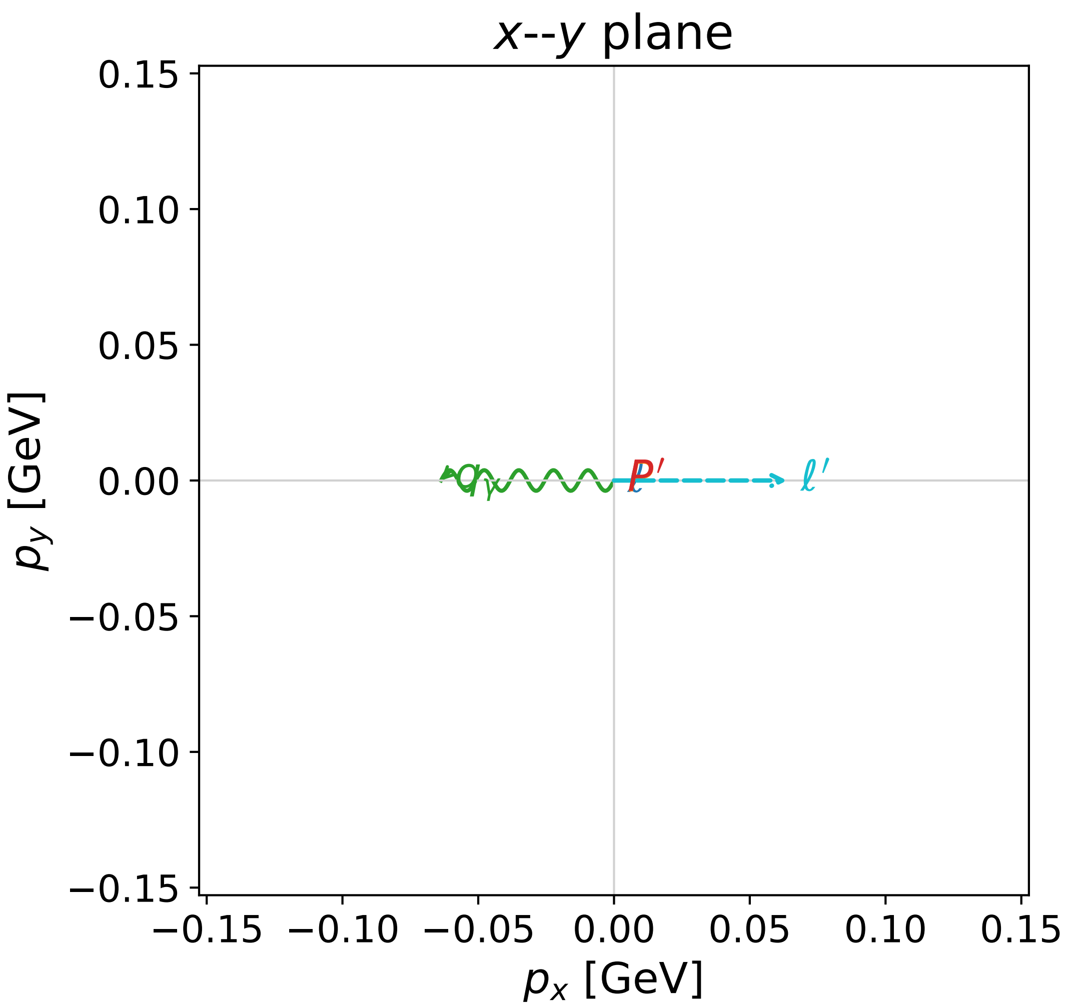


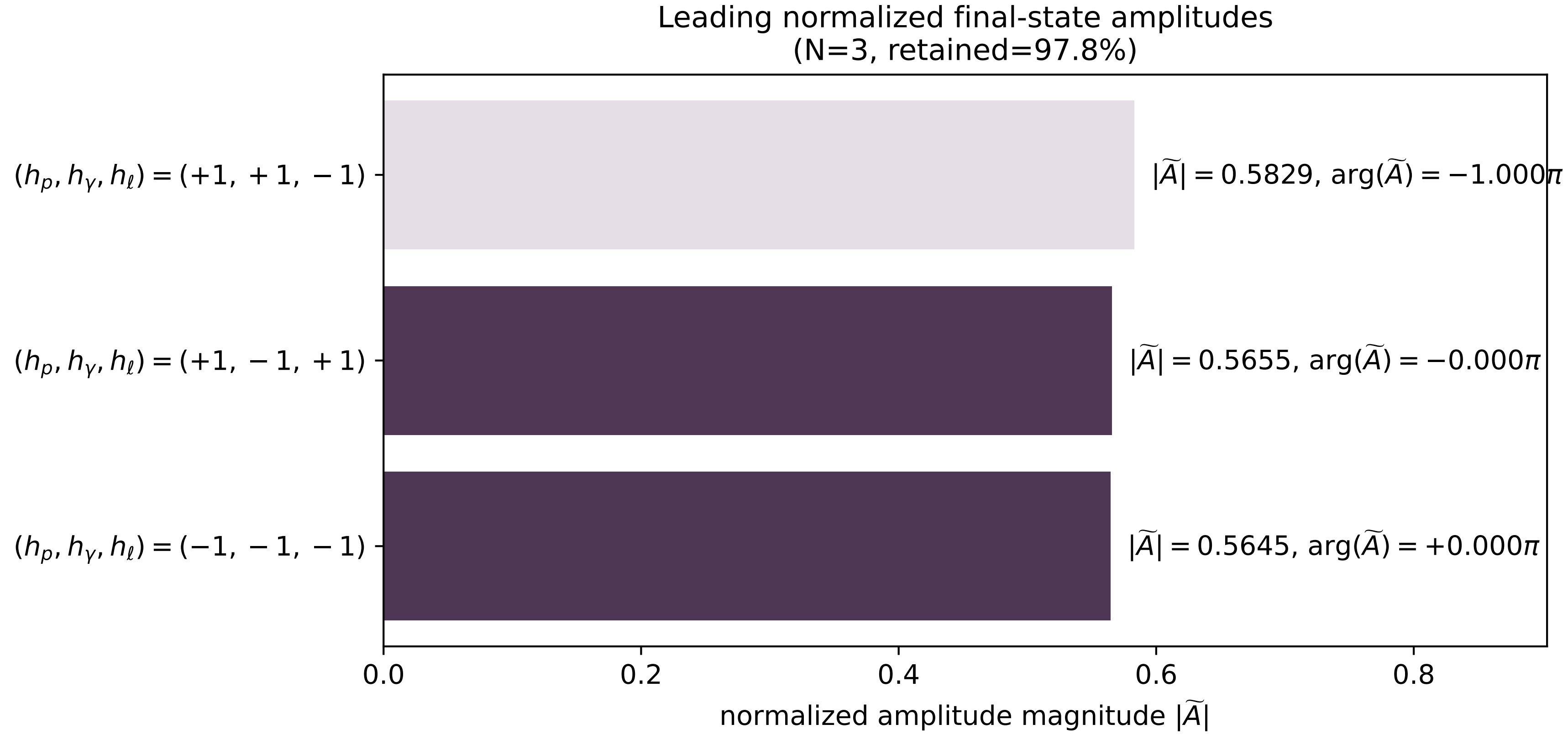

```
dw_mixing_angles_polarization_06_local_minimum_840
D_W=0.0116403
region: polarization_cluster_06_local_minimum_840
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0840
lepton mixing angle: alpha_e=2.3185035 rad
incoming lepton: -0.679959|+> +0.73325|->
proton mixing angle: alpha_p=1.7221857 rad
incoming proton: -0.150812|+> +0.988562|->
```

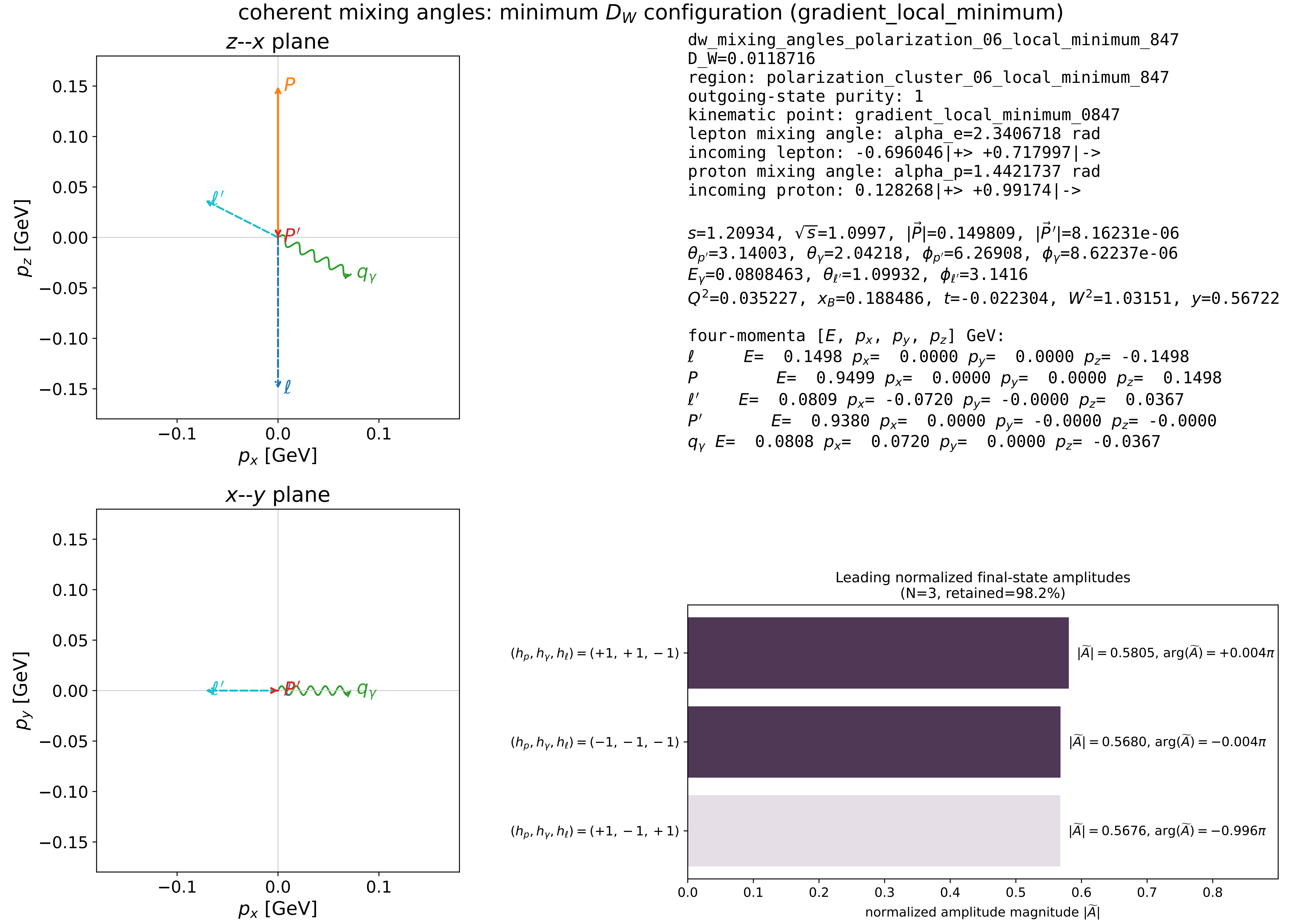
s=1.15333, $\sqrt{s}$=1.07393, $|\vec{P}|$=0.12733, $|\vec{P}'|$=0.0294481
$\theta_{p'}$=3.14159, θ_γ=1.62221, $\phi_{p'}$=4.91412, ϕ_γ=3.14161
E_γ=0.0638214, $\theta_{\ell'}$=1.09642, $\phi_{\ell'}$=1.27813e-05
Q^2=0.0265808, x_B=0.181842, t=-0.0245132, W^2=0.999439, y=0.534482

four-momenta [E, p_x, p_y, p_z] GeV:

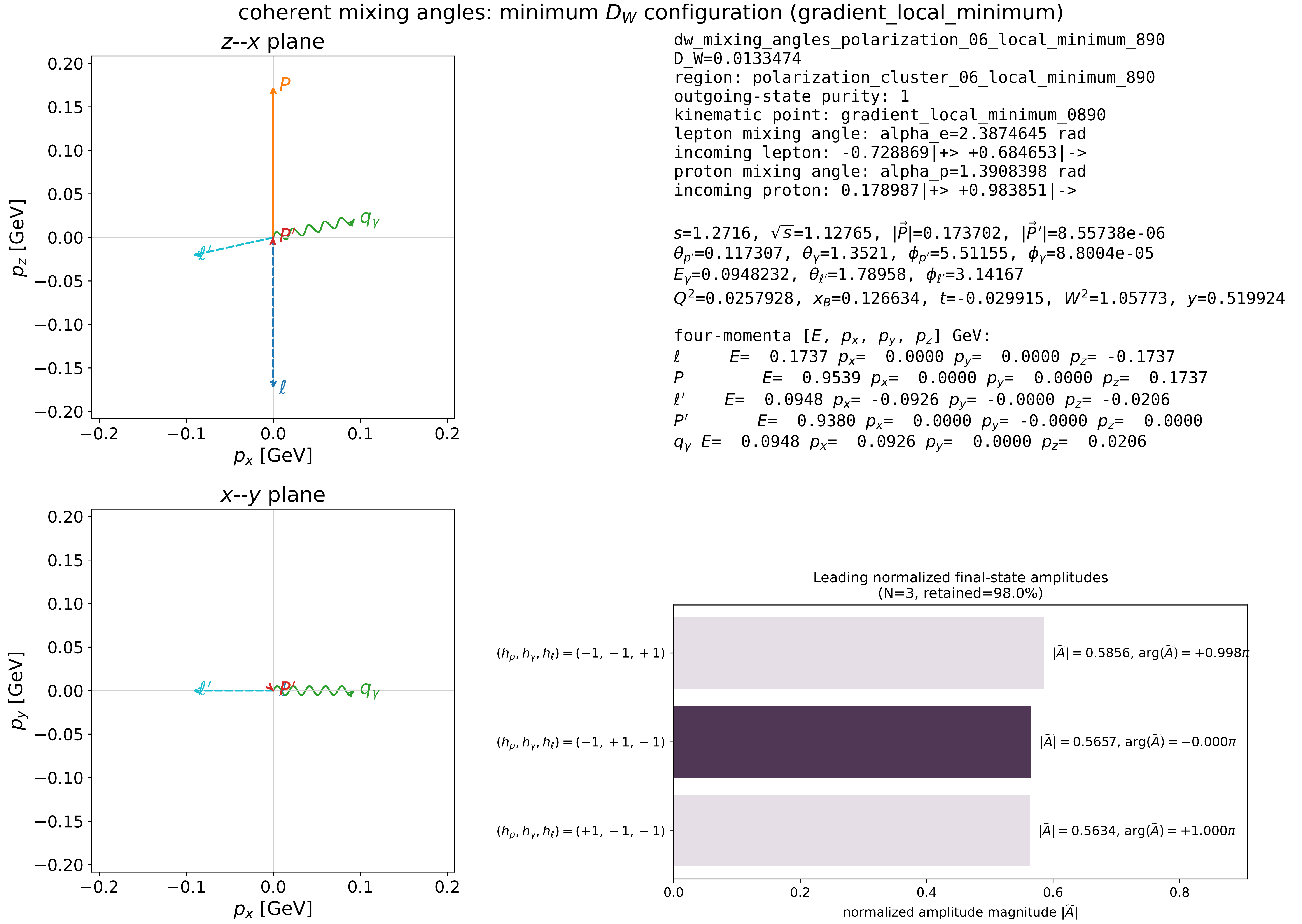
ℓ	E= 0.1273	p_x= 0.0000	p_y= 0.0000	p_z= -0.1273
P	E= 0.9466	p_x= 0.0000	p_y= 0.0000	p_z= 0.1273
ℓ'	E= 0.0717	p_x= 0.0637	p_y= 0.0000	p_z= 0.0327
P'	E= 0.9385	p_x= 0.0000	p_y= -0.0000	p_z= -0.0294
q_γ	E= 0.0638	p_x= -0.0637	p_y= -0.0000	p_z= -0.0033



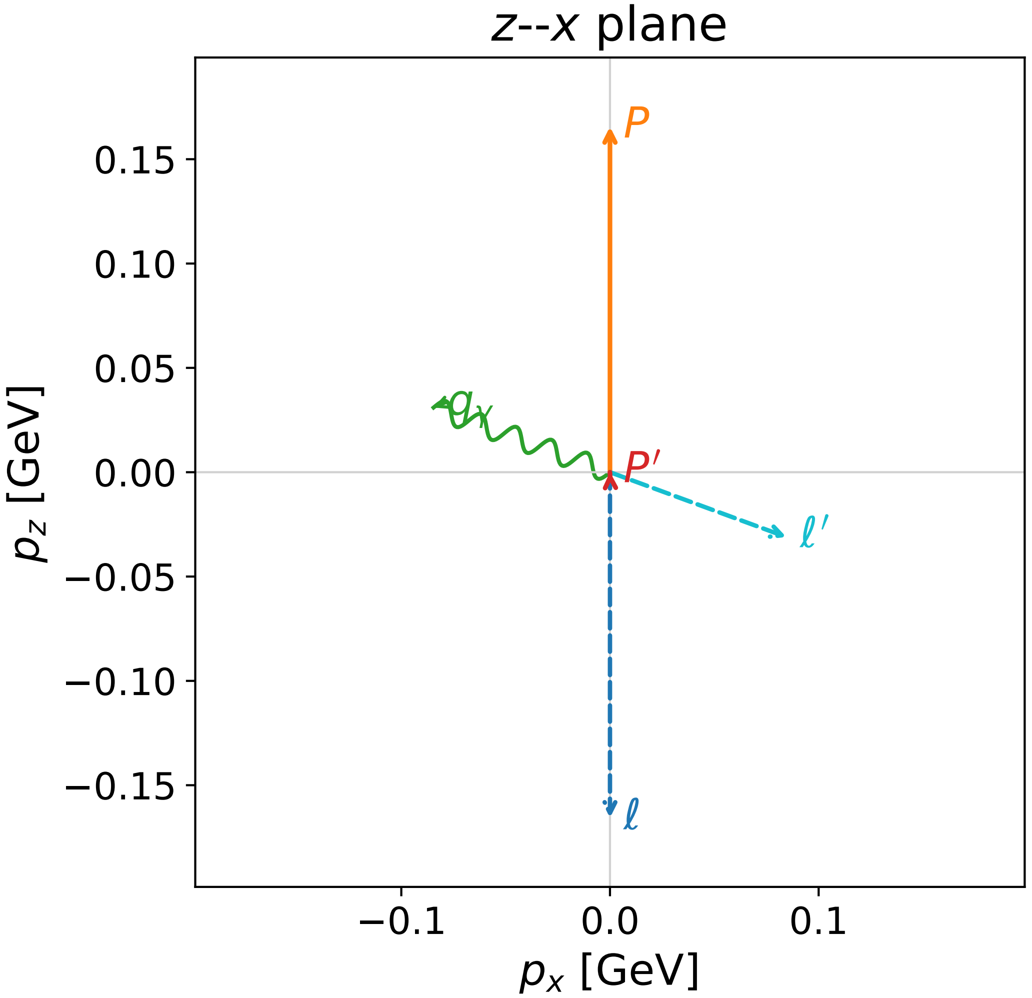




coherent mixing angles: minimum D_W configuration (gradient_local_minimum)
z--x plane
x--y plane
p_z [GeV]
p_y [GeV]
p_x [GeV]
P
P'
ℓ
ℓ'
q_γ
dw_mixing_angles_polarization_06_local_minimum_847
D_W=0.0118716
region: polarization_cluster_06_local_minimum_847
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0847
lepton mixing angle: alpha_e=2.3406718 rad
incoming lepton: -0.696046|+> +0.717997|->
proton mixing angle: alpha_p=1.4421737 rad
incoming proton: 0.128268|+> +0.99174|->
s=1.20934, $\sqrt{s}$=1.0997, $|\vec{P}|$=0.149809, $|\vec{P}'|$=8.16231e-06
$\theta_{p'}$=3.14003, θ_γ=2.04218, $\phi_{p'}$=6.26908, ϕ_γ=8.62237e-06
E_γ=0.0808463, $\theta_{\ell'}$=1.09932, $\phi_{\ell'}$=3.1416
Q^2=0.035227, x_B=0.188486, t=-0.022304, W^2=1.03151, y=0.56722
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1498 p_x= 0.0000 p_y= 0.0000 p_z= -0.1498
P E= 0.9499 p_x= 0.0000 p_y= 0.0000 p_z= 0.1498
ℓ' E= 0.0809 p_x= -0.0720 p_y= -0.0000 p_z= 0.0367
P' E= 0.9380 p_x= 0.0000 p_y= -0.0000 p_z= -0.0000
q_γ E= 0.0808 p_x= 0.0720 p_y= 0.0000 p_z= -0.0367
Leading normalized final-state amplitudes
(N=3, retained=98.2%)
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$
$|\widetilde{A}| = 0.5805$, $\arg(\widetilde{A}) = +0.004\pi$
$(h_p, h_\gamma, h_\ell) = (-1, -1, -1)$
$|\widetilde{A}| = 0.5680$, $\arg(\widetilde{A}) = -0.004\pi$
$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$
$|\widetilde{A}| = 0.5676$, $\arg(\widetilde{A}) = -0.996\pi$
normalized amplitude magnitude $|\widetilde{A}|$


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)
z--x plane
x--y plane
p_x [GeV]
p_z [GeV]
p_y [GeV]
P
P'
q_γ
ℓ'
ℓ
dw_mixing_angles_polarization_06_local_minimum_890
D_W=0.0133474
region: polarization_cluster_06_local_minimum_890
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0890
lepton mixing angle: alpha_e=2.3874645 rad
incoming lepton: -0.728869|+> +0.684653|->
proton mixing angle: alpha_p=1.3908398 rad
incoming proton: 0.178987|+> +0.983851|->
s=1.2716, $\sqrt{s}$=1.12765, $|\vec{P}|$=0.173702, $|\vec{P}'|$=8.55738e-06
$\theta_{p'}$=0.117307, θ_γ=1.3521, $\phi_{p'}$=5.51155, ϕ_γ=8.8004e-05
E_γ=0.0948232, $\theta_{\ell'}$=1.78958, $\phi_{\ell'}$=3.14167
Q^2=0.0257928, x_B=0.126634, t=-0.029915, W^2=1.05773, y=0.519924
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1737 p_x= 0.0000 p_y= 0.0000 p_z= -0.1737
P E= 0.9539 p_x= 0.0000 p_y= 0.0000 p_z= 0.1737
ℓ' E= 0.0948 p_x= -0.0926 p_y= -0.0000 p_z= -0.0206
P' E= 0.9380 p_x= 0.0000 p_y= -0.0000 p_z= 0.0000
q_γ E= 0.0948 p_x= 0.0926 p_y= 0.0000 p_z= 0.0206
Leading normalized final-state amplitudes
(N=3, retained=98.0%)
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$
$|\widetilde{A}| = 0.5856$, $\arg(\widetilde{A}) = +0.998\pi$
$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$
$|\widetilde{A}| = 0.5657$, $\arg(\widetilde{A}) = -0.000\pi$
$(h_p, h_\gamma, h_\ell) = (+1, -1, -1)$
$|\widetilde{A}| = 0.5634$, $\arg(\widetilde{A}) = +1.000\pi$
normalized amplitude magnitude $|\widetilde{A}|$

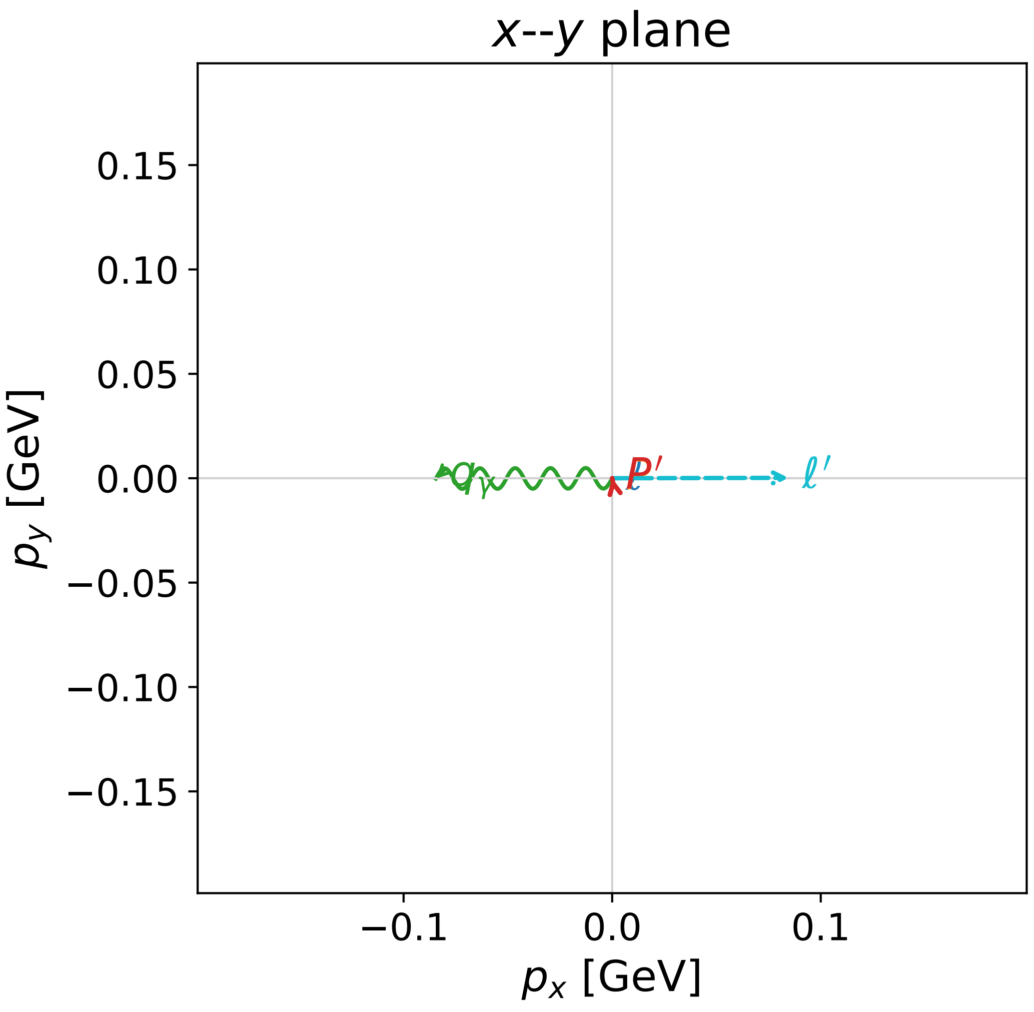
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

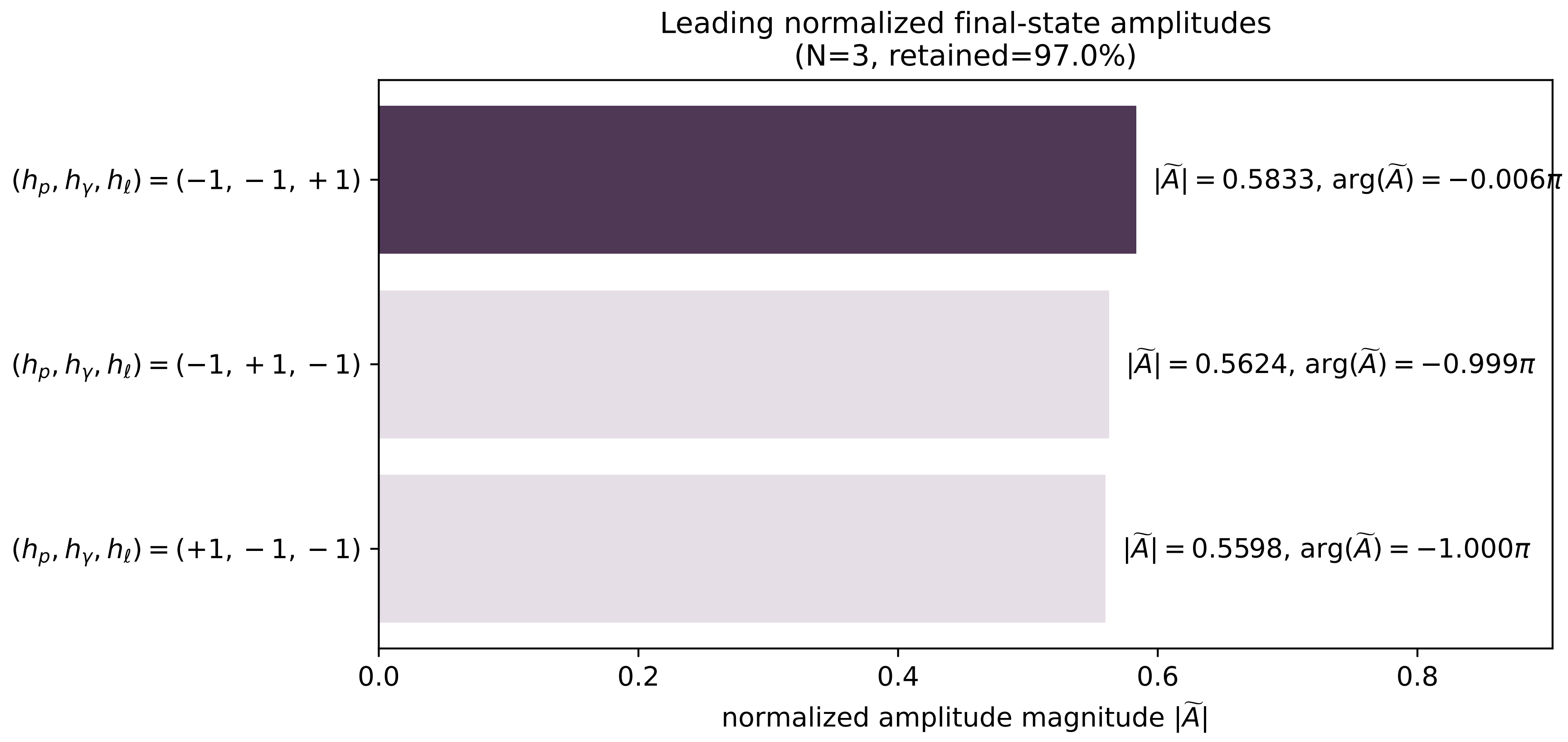


dw_mixing_angles_polarization_06_local_minimum_930
D_W=0.0148952
region: polarization_cluster_06_local_minimum_930
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0930
lepton mixing angle: alpha_e=2.3871059 rad
incoming lepton: -0.728623|+> +0.684915|->
proton mixing angle: alpha_p=1.728226 rad
incoming proton: -0.15678|+> +0.987634|->

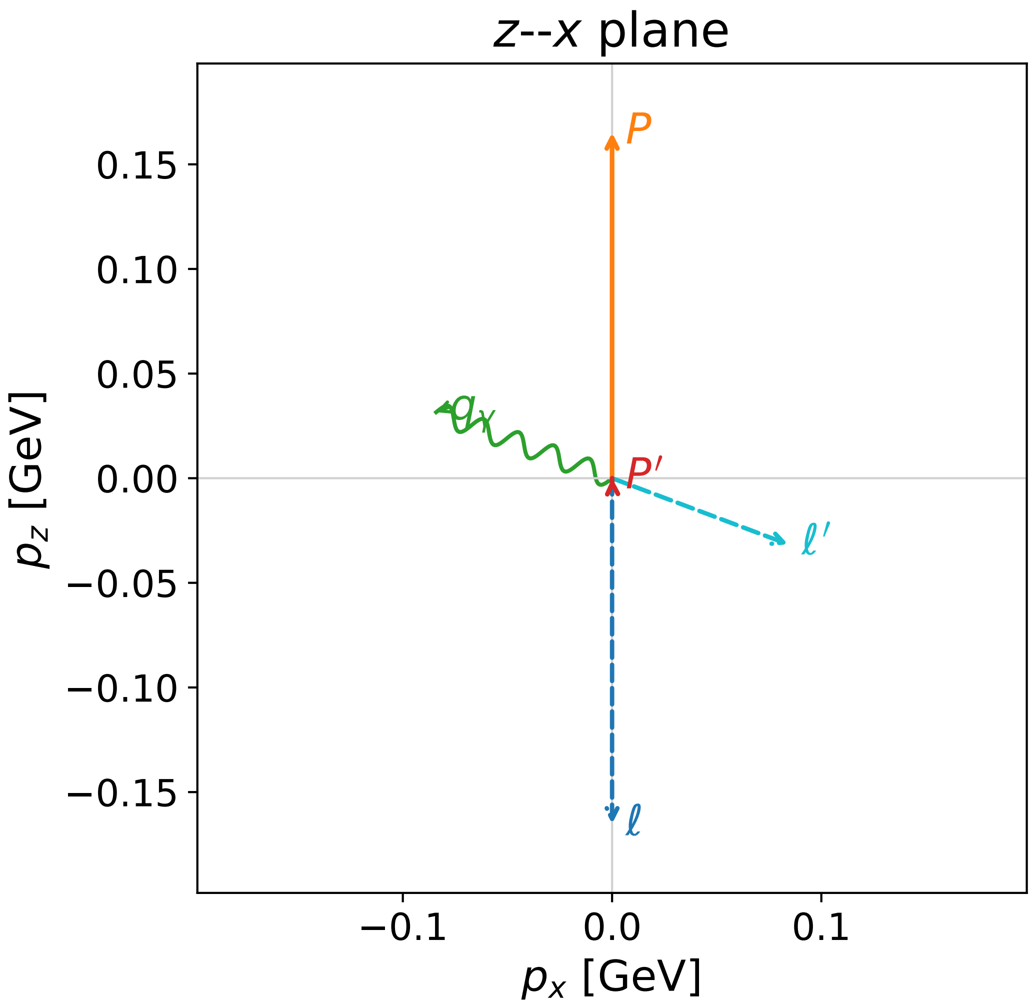
s=1.25021, $\sqrt{s}$=1.11813, $|\vec{P}|$=0.165617, $|\vec{P}'|$=8.61277e-05
$\theta_{p'}$=0.166897, θ_γ=1.21964, $\phi_{p'}$=1.75585, ϕ_γ=3.14396
E_γ=0.0900467, $\theta_{\ell'}$=1.92283, $\phi_{\ell'}$=0.00219716
Q^2=0.0195491, x_B=0.103724, t=-0.0271904, W^2=1.04877, y=0.508882

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1656 p_x= 0.0000 p_y= 0.0000 p_z= -0.1656
P E= 0.9525 p_x= 0.0000 p_y= 0.0000 p_z= 0.1656
ℓ' E= 0.0901 p_x= 0.0846 p_y= 0.0002 p_z= -0.0311
P' E= 0.9380 p_x= -0.0000 p_y= 0.0000 p_z= 0.0001
q_γ E= 0.0900 p_x= -0.0846 p_y= -0.0002 p_z= 0.0310





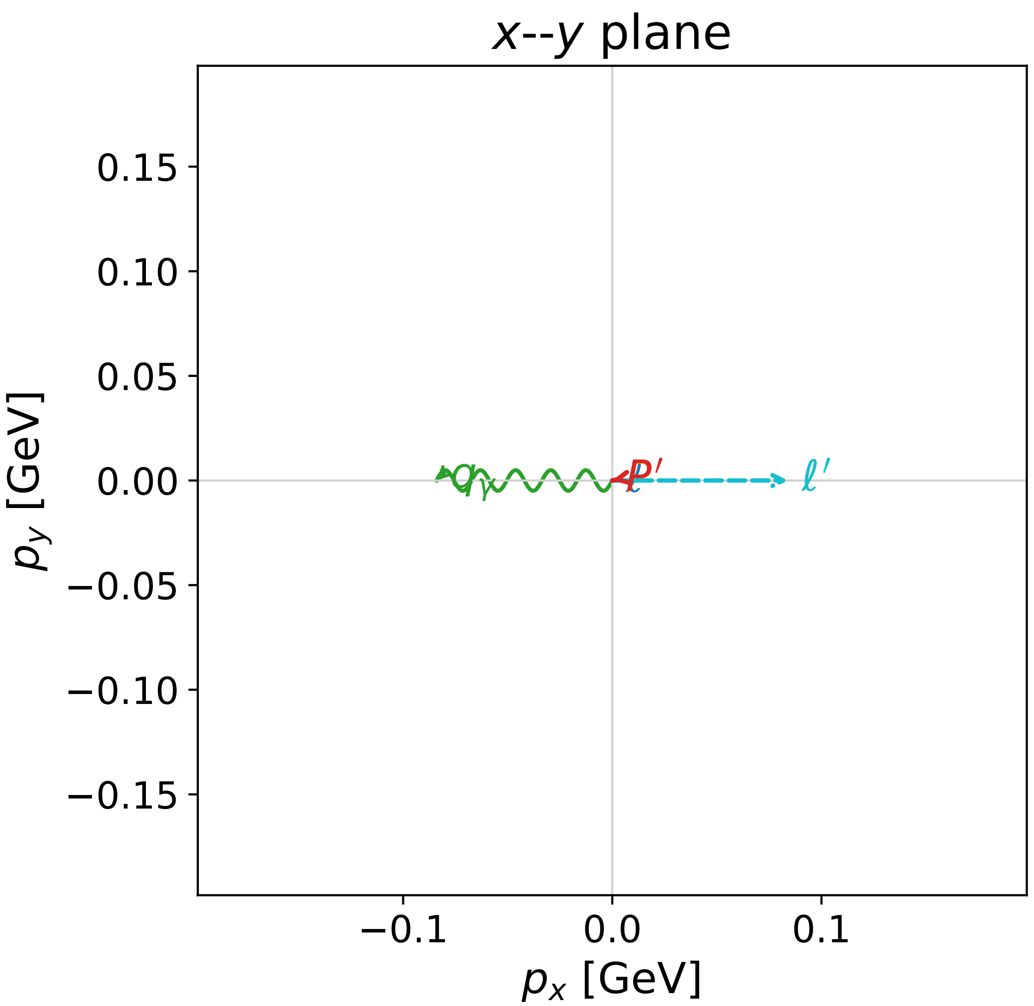
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

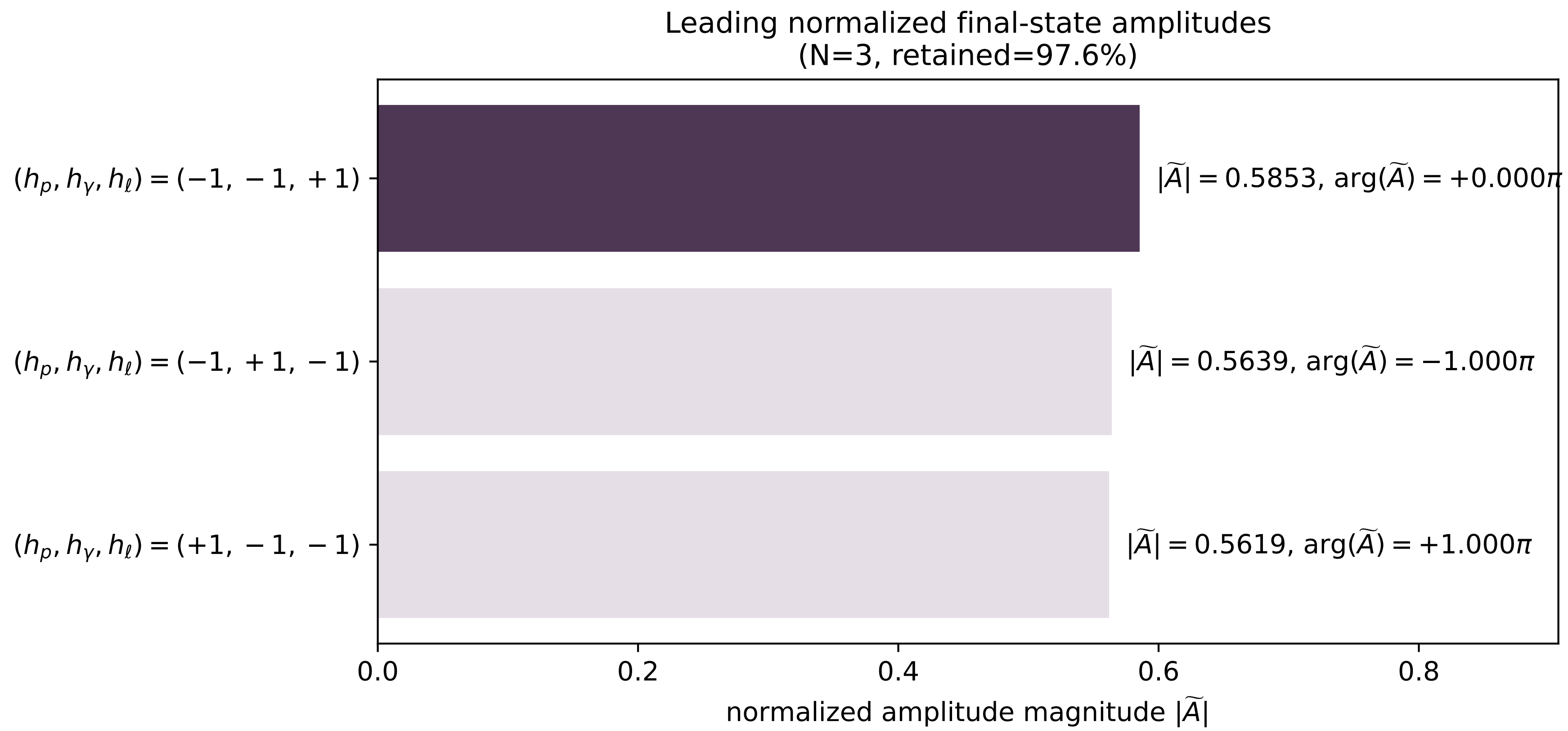



```
dw_mixing_angles_polarization_06_local_minimum_932
D_W=0.0150586
region: polarization_cluster_06_local_minimum_932
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0932
lepton mixing angle: alpha_e=2.3873934 rad
incoming lepton: -0.72882|+> +0.684705|->
proton mixing angle: alpha_p=1.7256129 rad
incoming proton: -0.154199|+> +0.98804|->
```

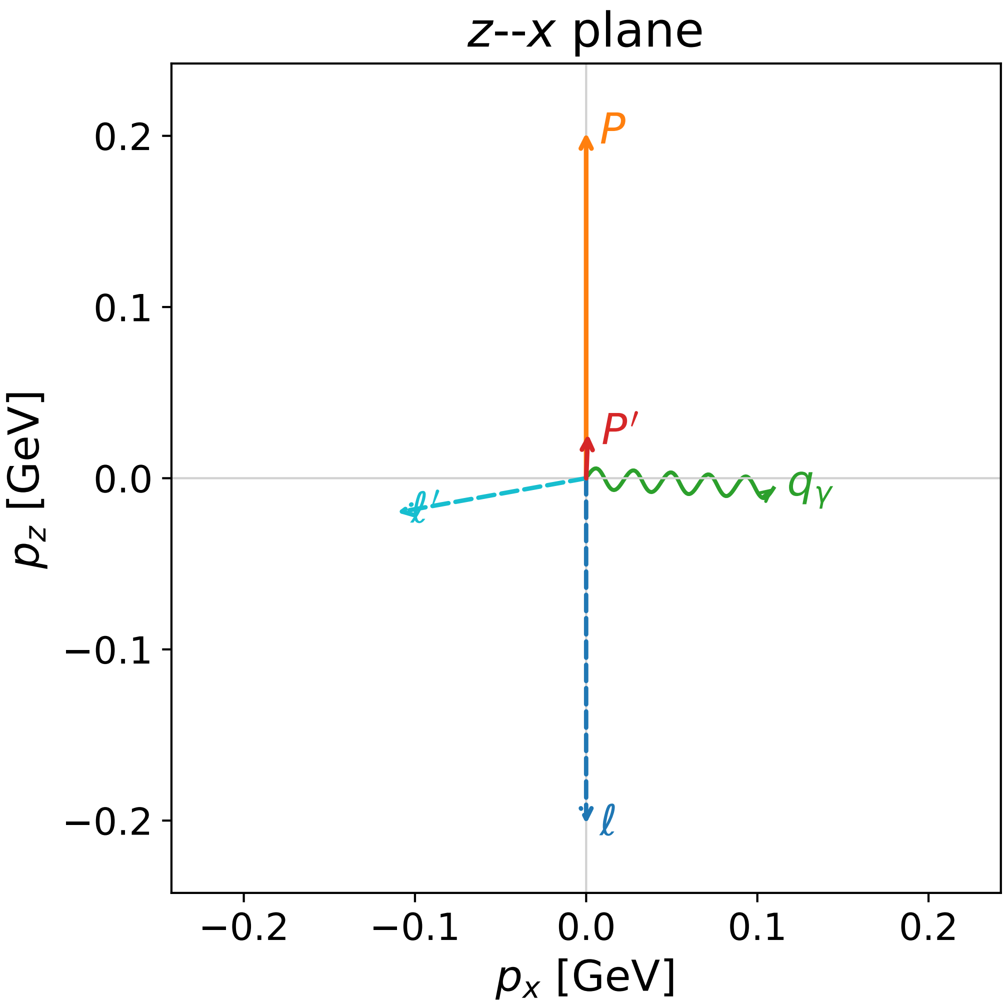
s=1.249, $\sqrt{s}$=1.11758, $|\vec{P}|$=0.165155, $|\vec{P}'|$=1.2777e-07
$\theta_{p'}$=0.0340212, θ_γ=1.21186, $\phi_{p'}$=3.33858, ϕ_γ=3.14161
E_γ=0.0897916, $\theta_{\ell'}$=1.92973, $\phi_{\ell'}$=1.74269e-05
Q^2=0.0192406, x_B=0.102513, t=-0.027068, W^2=1.04829, y=0.508435

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1652 p_x= 0.0000 p_y= 0.0000 p_z= -0.1652
P E= 0.9524 p_x= 0.0000 p_y= 0.0000 p_z= 0.1652
ℓ' E= 0.0898 p_x= 0.0841 p_y= 0.0000 p_z= -0.0315
P' E= 0.9380 p_x= -0.0000 p_y= -0.0000 p_z= 0.0000
q_γ E= 0.0898 p_x= -0.0841 p_y= -0.0000 p_z= 0.0315





coherent mixing angles: minimum D_W configuration (gradient_local_minimum)



dw_mixing_angles_polarization_06_local_minimum_933
D_W=0.0151282
region: polarization_cluster_06_local_minimum_933
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0933
lepton mixing angle: alpha_e=2.389794 rad
incoming lepton: -0.730462|+> +0.682954|->
proton mixing angle: alpha_p=1.3723144 rad
incoming proton: 0.197181|+> +0.980367|->

s=1.34877, $\sqrt{s}$=1.16137, $|\vec{P}|$=0.201885, $|\vec{P}'|$=0.0259237
$\theta_{p'}$=0.0702731, θ_γ=1.62406, $\phi_{p'}$=0.988205, ϕ_γ=6.17259
E_γ=0.110269, $\theta_{\ell'}$=1.74905, $\phi_{\ell'}$=3.04561
Q^2=0.0374493, x_B=0.153158, t=-0.0305422, W^2=1.08691, y=0.521433

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.2019 p_x= 0.0000 p_y= 0.0000 p_z= -0.2019
P E= 0.9595 p_x= 0.0000 p_y= 0.0000 p_z= 0.2019
ℓ' E= 0.1127 p_x= -0.1104 p_y= 0.0106 p_z= -0.0200
P' E= 0.9384 p_x= 0.0010 p_y= 0.0015 p_z= 0.0259
q_γ E= 0.1103 p_x= 0.1094 p_y= -0.0122 p_z= -0.0059